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The Singularity Is Near: When Humans Transcend Biology

by Ray Kurzweil

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638 Highlights | 12 Notes

Highlight (Yellow) | Location 252

“many paths to the truth.”

Highlight (Yellow) | Location 254

basic truths were profound enough to transcend apparent contradictions.

Highlight (Yellow) | Location 279

Even though I was quite familiar with every line of code, it nonetheless seemed as if the computer were deep in thought when the lights dimmed for several seconds at the denouement of each such cycle. Indeed, it could do flawlessly in ten seconds what took us ten hours to do manually with far less accuracy. As an inventor in the 1970s, I came to realize that my inventions needed to make sense in terms of the enabling technologies and market forces that would exist when the inventions were introduced, as that world would be a very different one from the one in which they were conceived.

Highlight (Yellow) | Location 315

In my teen years, I replaced my parlor magic with technology projects. I discovered that unlike mere tricks, technology does not lose its transcendent power when its secrets are revealed. I am often reminded of Arthur C. Clarke’s third law, that “any sufficiently advanced technology is indistinguishable from magic.”

Highlight (Yellow) | Location 339

the destiny of the human-machine civilization, a destiny we have come to refer to as the Singularity.

Highlight (Yellow) | Location 356

and I still cannot say that I am entirely comfortable with all of its consequences.

Highlight (Yellow) | Location 364

the last seven doublings were sufficient to extend the pads' coverage to the entire lake. (Seven doublings extended their reach 128-fold.) This is the nature of exponential growth. Consider Gary Kasparov, who scorned the pathetic state of computer chess in 1992. Yet the relentless doubling of computer power every year enabled a computer to defeat him only five years later.²

Highlight (Yellow) | Location 377

our thinking is extremely slow: the basic neural transactions are several million times slower than contemporary electronic circuits. That makes our physiological bandwidth for processing new information extremely limited compared to the exponential growth of the overall human knowledge base.

Highlight (Yellow) | Location 386

beginning to reach the "knee of the curve," which is the stage at which an exponential trend becomes noticeable.

Highlight (Yellow) | Location 399

When the first transhuman intelligence is created and launches itself into recursive self-improvement, a fundamental discontinuity is likely to occur, the likes of which I can't even begin to predict. —MICHAEL ANISSIMOV

Highlight (Yellow) | Location 420

in the twenty-first century; we will witness on the order of twenty thousand years of progress (again, when measured by today's rate of progress), or about one thousand times greater than what was achieved in the twentieth century.

Highlight (Yellow) | Location 472

nanotechnology-based manufacturing devices in the 2020s will be capable of creating almost any physical product from inexpensive raw materials and information.

Highlight (Yellow) | Location 477

Six Epochs First we build the tools, then they build us. —MARSHALL MCLUHAN The future ain't what it used to be. —YOGI BERRA

Highlight (Yellow) | Location 482

patterns of increasing order. I'll discuss the concept of order in the next chapter; the emphasis in this section is on the concept of patterns.

Highlight (Yellow) | Location 483

Evolution works through indirection: each stage or epoch uses the information-processing methods of the previous epoch to create the next.

Highlight (Orange) | Location 486

Epoch One: Physics and Chemistry.

Highlight (Orange) | Location 504

Epoch Two: Biology and DNA.

Highlight (Orange) | Location 508

Epoch Three: Brains.

Highlight (Yellow) | Location 512

third epoch started with the ability of early animals to recognize patterns, which still accounts for the vast majority of the activity in our brains.

Highlight (Orange) | Location 516

Epoch Four: Technology.

Highlight (Yellow) | Location 552

mass adoption (meaning that they are used by a quarter of the population in advanced countries)

Highlight (Orange) | Location 553

Epoch Five: The Merger of Human Technology with Human Intelligence.

Highlight (Yellow) | Location 555

The fifth epoch will enable our human-machine civilization to transcend the human brain's limitations of a mere hundred trillion extremely slow connections.¹⁴ The Singularity

Highlight (Orange) | Location 559

Epoch Six: The Universe Wakes Up.

Highlight (Yellow) | Location 569

This is the ultimate destiny of the Singularity and of the universe.

Highlight (Yellow) | Location 625

I emphasize the word “currently” because one of the salient implications of the Singularity will be a change in the nature of our ability to understand. We will become vastly smarter as we merge with our technology.

Highlight (Yellow) | Location 632

When scientists become a million times more intelligent and operate a million times faster, an hour would result in a century of progress (in today’s terms).

Highlight (Yellow) | Location 649

We will have effective software models of human intelligence by the mid-2020s.

Highlight (Yellow) | Location 650

we can expect computers to pass the Turing test, indicating intelligence indistinguishable from that of biological humans, by the end of the 2020s.³⁰

Highlight (Yellow) | Location 653

of human intelligence include a formidable ability to recognize patterns.

Highlight (Yellow) | Location 655

Humans are also capable of learning new knowledge by applying insights and inferring principles from experience, including information gathered through language.

Highlight (Yellow) | Location 658

The traditional strengths of machine intelligence include the ability to remember billions of facts precisely and recall them instantly. • Another advantage of nonbiological intelligence is that once a skill is mastered by a machine, it can be performed repeatedly at high speed, at optimal accuracy, and without tiring. • Perhaps most important, machines can share their knowledge at extremely high speed,

Highlight (Yellow) | Location 662

Nonbiological intelligence will be able to download skills and knowledge from other machines, eventually also from humans.

Highlight (Yellow) | Location 663

Machines will process and switch signals at close to the speed of light (about three hundred million meters per second), compared to about one hundred meters per second for the electrochemical signals used in biological mammalian brains.³¹

Highlight (Yellow) | Location 666

Machines will have access via the Internet to all the knowledge

Highlight (Yellow) | Location 667

Machines can pool their resources, intelligence, and memories.

Highlight (Yellow) | Location 668

join together to become one and then become separate again. Multiple machines can do both at the same time: become one and separate simultaneously. Humans call this falling in love,

Highlight (Yellow) | Location 706

Nanobots will interact with biological neurons to vastly extend human experience by creating virtual reality from within the nervous system.

Highlight (Yellow) | Location 708

Billions of nanobots in the capillaries of the brain will also vastly extend human intelligence.

Highlight (Yellow) | Location 710

biological intelligence is effectively of fixed capacity. Thus, the nonbiological portion of our intelligence will ultimately predominate.

Highlight (Yellow) | Location 712

Nanobots will also enhance the environment by reversing pollution from earlier industrialization.

Highlight (Yellow) | Location 713

Nanobots called foglets that can manipulate image and sound waves will bring the morphing qualities of virtual reality to the real world.

Highlight (Yellow) | Location 723

In fact, other people (such as your romantic partner) will be able to select a different body for you than you might select for yourself (and vice versa).

Highlight (Yellow) | Location 738

Most important, the intelligence that will emerge will continue to represent the human civilization, which is already a human-machine civilization. In other words, future machines will be human, even if they are not biological. This will be the next step in evolution,

Highlight (Yellow) | Location 822

RAY: Still, you represent only a tiny fraction of the available matter and energy. Keeping you biological won't appreciably change the order of magnitude of matter and energy available to the Singularity. It will be well worth it to maintain the biological heritage. GEORGE 2048: Absolutely. RAY: Just like today we seek to preserve the rain forest and the diversity of species.

Highlight (Yellow) | Location 864

Our sole responsibility is to produce something smarter than we are; any problems beyond that are not ours to solve

Highlight (Yellow) | Location 979

Technology will be the metaphorical opposable thumb that enables our next step in evolution. Progress (further increases in order) will then be based on thinking processes that occur at the speed of light rather than in very slow electrochemical reactions. Each stage of evolution builds on the fruits of the last stage,

Highlight (Yellow) | Location 1012

Indeed, the emergence of the first technology-creating species resulted in the new evolutionary process of technology, which makes technological evolution an outgrowth of—and a continuation of—biological evolution.

Highlight (Yellow) | Location 1018

A specific paradigm (a method or approach to solving a problem; for example, shrinking transistors on an integrated circuit as a way to make more powerful computers) generates exponential growth until its potential is exhausted. When this happens, a paradigm shift occurs, which enables exponential growth to continue.

Highlight (Yellow) | Location 1078

we need to understand its design principles, which are far simpler (that is, contain far less information) than the extremely detailed structures that the genetic information generates through these iterative, fractal-like processes.

Highlight (Yellow) | Location 1082

Consider how the principles of the law of accelerating returns apply to the epochs we discussed in the first chapter. The combination of amino acids into proteins and of nucleic acids into strings of RNA established the basic paradigm of biology. Strings of RNA (and later DNA) that self-replicated (Epoch Two) provided a digital method to record the results of evolutionary experiments. Later on, the evolution of a species that combined rational thought (Epoch Three) with an opposable appendage (the thumb) caused a fundamental paradigm shift from biology to technology (Epoch Four). The upcoming primary paradigm shift will be from biological thinking to a hybrid combining biological and nonbiological thinking (Epoch Five), which will include “biologically inspired” processes resulting from the reverse engineering of biological brains. If we examine the timing of these epochs, we see that they have been part of a continuously accelerating process. The evolution of life-forms required billions of years for its first steps (primitive cells, DNA), and then progress accelerated. During the Cambrian

Highlight (Yellow) | Location 1096

Consider the boundaries of observation. Early biological life could observe local events several millimeters away, using chemical gradients. When sighted animals evolved, they were able to observe events that were miles away. With the invention of the telescope, humans could see other galaxies millions of light-years away. Conversely, using microscopes, they could also see cellular-size structures. Today humans armed with contemporary technology can see to the edge of the observable universe, a distance of more than thirteen billion light-years, and down to quantum-scale subatomic particles.

Highlight (Yellow) | Location 1127

The Life Cycle of a Technology We can identify seven distinct stages in the life cycle of a technology.

Highlight (Blue) | Location 1130

dreaming

Highlight (Blue) | Location 1132

invention,

Highlight (Blue) | Location 1135

protected and supported by doting guardians

Highlight (Yellow) | Location 1137

but it was Henry Ford’s innovation of mass production that enabled the automobile to take root and flourish.

Highlight (Blue) | Location 1138

maturity.

Highlight (Blue) | Location 1141

upstart threatens to eclipse

Highlight (Blue) | Location 1145

obsolescence.

Highlight (Blue) | Location 1147

yields to antiquity

Highlight (Yellow) | Location 1163

The false pretender was the electric piano of the early 1980s.

Highlight (Yellow) | Location 1190

As with the phonograph record and the piano, this first generation of false pretenders was (and still is) missing an essential quality of the original, which in this case is the superb visual characteristics of paper and ink. Paper does not flicker, whereas the typical computer screen is displaying sixty or more fields per second. This is a problem because of an evolutionary adaptation of the primate visual system. We are able to see only a very small portion of the visual field with high resolution. This portion, imaged by the fovea in the retina, is focused on an area about the size of a single word at twenty-two inches away. Outside of the fovea, we have very little resolution but exquisite sensitivity to changes in brightness, an ability that allowed our primitive forebears to quickly detect a predator that might be attacking. The constant flicker of a video graphics array (VGA) computer screen is detected by our eyes as motion and causes constant movement of the fovea. This substantially slows down reading speeds, which is one reason that reading on a screen is less pleasant than reading a printed book. This particular

Highlight (Yellow) | Location 1217

Before considering further the implications of the Singularity, let's examine the wide range of technologies that are subject to the law of accelerating returns. The exponential trend that has gained the greatest public recognition has become known as Moore's Law.

Highlight (Yellow) | Location 1308

If we project these computational performance trends through this next century, we can see in the figure below that supercomputers will match human brain capability by the end of this decade and personal computing will achieve it by around 2020—or possibly sooner,

Highlight (Yellow) | Location 1312

The exponential growth of computing is a marvelous quantitative example of the exponentially growing returns from an evolutionary process. We can express the exponential growth of computing in terms of its accelerating pace: it took ninety years to achieve the first MIPS per thousand dollars; now we add one MIPS per thousand dollars every five hours.

Highlight (Yellow) | Location 1316

one million gigaflops (billions of floating-point operations per second), or 10^{15} calculations per second when it launches in 2007.⁴¹ That's one tenth of the 10^{16} calculations per second needed to emulate the human brain

Highlight (Yellow) | Location 1324

human brain uses a very inefficient electrochemical, digital-controlled analog computational process. The bulk of its calculations are carried out in the interneuronal connections at a speed of only about two hundred calculations per second (in each connection), which is at least one million times slower than contemporary electronic circuits.

Highlight (Yellow) | Location 1329

we won't approach those limits until late in this century. It is important to distinguish between the S-curve that is characteristic of any specific technological paradigm and the continuing exponential growth that is characteristic of the ongoing evolutionary process within a broad area of technology, such as computation.

Highlight (Yellow) | Location 1342

Things are more like they are now than they ever were before. —DWIGHT D. EISENHOWER

Highlight (Yellow) | Location 1359

A dramatic recent example of this improving capacity was the sequencing of the SARS virus, which took only thirty-one days from the identification of the virus, compared to more than fifteen years for the HIV virus.⁴⁶

Highlight (Yellow) | Location 1375

As has been discussed, while technology progresses exponentially, we experience it in the linear domain.

Highlight (Yellow) | Location 1461

The discovery that has so excited Wolfram is a simple rule he calls cellular automata rule 110 and its behavior. (There are some other interesting automata rules, but rule 110

Highlight (Yellow) | Location 1484

Wolfram makes the following point repeatedly: “Whenever a phenomenon is encountered that seems complex it is taken almost for granted that the phenomenon must be the result of some underlying mechanism that is itself complex. But my discovery that simple programs can produce great complexity makes it clear that this is not in fact correct.”⁶⁷ I do find the behavior of rule 110 rather delightful. Furthermore, the idea that a completely deterministic process can produce results that are completely unpredictable is of great importance, as it provides an explanation for how the world can be inherently unpredictable while still based on fully deterministic rules.⁶⁸

Highlight (Yellow) | Location 1559

that has already taken place. We will be applying evolutionary algorithms within these solutions just as the human brain does. For example, the fetal wiring is initially random within constraints specified in the genome in at least some regions. Recent research shows that areas having to do with learning undergo more change, whereas structures having to do with sensory processing experience less change after birth.

Highlight (Yellow) | Location 1564

Given that the universe runs at the fastest speed it can run, there is usually no way to short-circuit the process. However, we have the benefits of the billions of years of evolution that have already taken place, which are responsible for the greatly increased order of complexity in the natural world. We can now benefit from it by using our evolved tools to reverse engineer the products of biological evolution (most importantly, the human brain).

Highlight (Yellow) | Location 1575

Charles Babbage’s late-nineteenth-century mechanical computer (which never ran) provided only a handful of operation codes yet provided (within its memory capacity and speed) the same kinds of transformations that modern computers do. The complexity of Babbage’s invention stemmed only from the details of its design, which indeed proved too difficult for Babbage to implement using the technology available to him. The Turing machine, Alan Turing’s theoretical conception of a universal computer in 1950, provides only seven very basic commands, yet can be organized to perform any possible computation.

Highlight (Yellow) | Location 1602

We have societies in Asia that jumped from agrarian economies to information economies, without going through industrialization.⁷⁷

Highlight (Yellow) | Location 1622

The reasonable man adapts himself to the world; the unreasonable one persists in trying to adapt the world to himself. Therefore all progress depends on the unreasonable man. —GEORGE BERNARD SHAW, “MAXIMS FOR REVOLUTIONISTS,” MAN AND SUPERMAN, 1903

Highlight (Yellow) | Location 1662

2042, approximately the time frame I've estimated for the Singularity (see the next chapter).

Highlight (Yellow) | Location 1836

As I discuss in *Engines of Creation*, if you can build genuine AI, there are reasons to believe that you can build things like neurons that are a million times faster. That leads to the conclusion that you can make systems that think a million times faster than a person. With AI, these systems could do engineering design. Combining this with the capability of a system to build something that is better than it, you have the possibility for a very abrupt transition.

Highlight (Yellow) | Location 1882

(NTT) has demonstrated a dramatic 3-D technology using electron-beam lithography, which can create arbitrary three-dimensional structures with feature sizes (such as transistors) as small as ten nanometers.⁶ NTT demonstrated the technology by creating a high-resolution model of the Earth sixty microns in size with ten-nanometer features.

Highlight (Yellow) | Location 1889

nanotubes are only one nanometer in diameter, so they can achieve high densities. They are also potentially very fast. Peter Burke and his colleagues at the University of California at Irvine recently demonstrated nanotube circuits operating at 2.5 gigahertz (GHz). However, in *Nano Letters*, a peer-reviewed journal of the American Chemical Society, Burke says the theoretical speed limit for these nanotube transistors "should be terahertz [1 THz = 1,000 GHz], which is about 1,000 times faster than modern computer speeds."⁸ One cubic inch of nanotube circuitry, once fully developed, would be up to one hundred million times more powerful than the human brain.⁹

Highlight (Yellow) | Location 1980

In 2003 Israeli scientists led by Ehud Shapiro at the Weizmann Institute of Science combined DNA with adenosine triphosphate (ATP), the natural fuel for biological systems such as the human body.²⁷ With this method, each of the DNA molecules was able to perform computations as well as provide its own energy. The Weizmann scientists demonstrated a configuration consisting of two spoonfuls of this liquid super-computing system, which contained thirty million billion molecular computers and performed a total of 660 trillion calculations per second (6.6×10^{14} cps). The energy consumption of these computers is extremely low, only fifty millionths of a watt for all thirty million billion computers. There's a limitation,

Highlight (Yellow) | Location 1986

many trillions of computers has to perform the same operation at the same time (although on different data), so that the device is a "single instruction multiple data" (SIMD) architecture.

Highlight (Yellow) | Location 1994

This concept is theoretical, because an electron is considered to occupy a point in space, so it is difficult to imagine a point with no size that nonetheless spins. However, when an electrical charge moves, it causes a magnetic field, which is real and measurable. An electron can spin in one of two directions, described as “up” and “down,” so this property can be exploited for logic switching or to encode a bit of memory.

Highlight (Yellow) | Location 2000

the spin of the electron can be transported without any loss of energy, or dissipation. Furthermore, this effect occurs at room temperature in materials already widely used in the semiconductor industry, such as gallium arsenide. That’s important because it could enable a new generation of computing devices.”²⁸ The potential, then, is to achieve the efficiencies of superconducting (that is, moving information at or close to the speed of light without any loss of information) at room temperature.

Highlight (Yellow) | Location 2014

An important role for spintronics in the future of computer memory is clear, and it is likely to contribute to logic systems as well. The spin of an electron is a quantum property (subject to the laws of quantum mechanics), so perhaps the most important application of spintronics will be in quantum computing systems, using the spin of quantum-entangled electrons to represent qubits, which I discuss below. Spin has also been used to store information in the nucleus of atoms, using the complex interaction of their protons’ magnetic moments. Scientists at the University of Oklahoma also demonstrated a “molecular photography” technique for storing 1,024 bits of information in a single liquid-crystal molecule comprising nineteen hydrogen atoms.³⁰

Highlight (Yellow) | Location 2026

general-purpose computing with its “multiple instruction multiple data” (MIMD) architectures

Highlight (Yellow) | Location 2071

program called EQP running five weeks on a 50 MIPS computer at Argonne National Laboratory found a proof of a Boolean algebra conjecture by Herbert Robbins that had eluded mathematicians for sixty years. And it is still only Spring. Wait until Summer. —HANS MORAVEC, “WHEN WILL COMPUTER HARDWARE MATCH THE HUMAN BRAIN?” 1997

Highlight (Yellow) | Location 2097

1014 (100 trillion) instructions per second for the entire brain.³⁷

Highlight (Yellow) | Location 2138

will achieve 1016 cps by 2025.

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Currently at least 99 percent, if not 99.9 percent, of the computational capacity of all the computers on the Internet lies unused. Effectively harnessing this computation can provide another factor of 102 or 103 in increased price-performance. For these reasons, it is reasonable to expect human brain capacity, at least in terms of hardware computational capacity, for one thousand dollars by around 2020.

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By 2050, one thousand dollars of computing will exceed the processing power of all human brains on Earth.

Highlight (Yellow) | Location 2183

Ultimately, we will be able to port our mental processes to a more suitable computational substrate. Then our minds won't have to stay so small.

Highlight (Yellow) | Location 2217

The current computer paradigm relies on what is known as irreversible computing, meaning that we are unable in principle to run software programs backward. At each step in the progression of a program, the input data is discarded—erased—and the results of the computation pass to the next step. Programs generally do not retain all intermediate results, as that would use up large amounts of memory unnecessarily. This selective erasure of input information is particularly true for pattern-recognition systems. Vision systems, for example, whether human or machine, receive very high rates of input (from the eyes or visual sensors) yet produce relatively compact outputs (such as identification of recognized patterns). This act of erasing data generates heat and therefore requires energy. When a bit of information is erased, that information has to go somewhere. According to the laws of thermodynamics, the erased bit is essentially released into the surrounding environment, thereby increasing its entropy, which can be viewed as a measure of information (including apparently disordered information) in an environment. This results in a higher temperature for the environment (because temperature is a measure of entropy). If, on the other hand, we don't erase each bit of information contained in the input to each step of an algorithm but instead just move it to another location, that bit stays in the computer, is not released into the environment, and therefore generates no heat and requires no energy from outside the computer.

Highlight (Yellow) | Location 2261

Fredkin's reversible logic gates answer a key challenge to the idea of reversible computing: that it would require a different style of programming. He argues that we can, in fact, construct normal logic and memory entirely from reversible logic gates, which will allow the use of existing conventional software-development methods. It is hard to overstate the significance of this insight. A key observation regarding the Singularity is that information processes—computation—will ultimately drive everything that is important. This primary foundation for future technology thus appears to require no energy.

Highlight (Yellow) | Location 2270

because of essentially random thermal and quantum effects, logic operations have an inherent error rate. We can overcome errors using error-detection and -correction codes, but each time we correct a bit, the operation is not reversible, which means it requires energy and generates heat.

Highlight (Yellow) | Location 2274

As we consider the limits of computation, the issue of error rate becomes a significant design issue. Certain methods of increasing computational rate, such as increasing the frequency of the oscillation of particles, also increase error rates,

Highlight (Yellow) | Location 2283

MIT professor Seth Lloyd has estimated the maximum computational capacity, according to the known laws of physics, of a computer weighing one kilogram and occupying one liter of volume—about the size and weight of a small laptop computer—what he calls the “ultimate laptop.”⁵⁸

Highlight (Yellow) | Location 2291

there is a direct proportional relationship between the energy of an object and its potential to perform computation.

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The potential of matter to compute is also governed by a very small number, Planck’s constant: 6.6×10^{-34} joule-seconds (a joule is a measure of energy). This is the smallest scale at which we can apply energy for computation. We obtain the theoretical limit of an object to perform computation by dividing the total energy (the average energy of each atom or particle times the number of such particles) by Planck’s constant.

Highlight (Yellow) | Location 2299

If we relate that figure to the most conservative estimate of human brain capacity (1019 cps and 1010 humans), it represents the equivalent of about five billion trillion human civilizations.⁶⁰ If we use the figure of 1016 cps that I believe will be sufficient for functional emulation of human intelligence, the ultimate laptop would function at the equivalent brain power of five trillion trillion human civilizations.⁶¹ Such a laptop could perform the equivalent of all human thought over the last ten thousand years (that is, ten billion human brains operating for ten thousand years) in one ten-thousandth of a nanosecond.⁶²

Highlight (Yellow) | Location 2305

a few caveats are in order. Converting all of the mass of our 2.2-pound laptop into energy is essentially what happens in a thermonuclear explosion.

Highlight (Yellow) | Location 2307

Lloyd shows that such a computer would have a theoretical memory capacity of 1031 bits. It's difficult to imagine technologies that would go all the way in achieving these limits. But we can readily envision technologies that come reasonably close to doing so. As the University of Oklahoma project shows, we already demonstrated the ability to store at least fifty bits of information per atom (although only on a small number of atoms, so far). Storing 1027 bits of memory in the 1025 atoms in a kilogram of matter should therefore be eventually achievable.

Highlight (Yellow) | Location 2314

we'll use the more conservative estimate.⁶³ As discussed above, 1042 calculations per second could be achieved

Highlight (Yellow) | Location 2316

By fully deploying reversible computing techniques, using designs that generate low levels of errors, and allowing for reasonable amounts of energy dissipation, we should end up somewhere between 1042 and 1050 calculations per second.

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Even at 1042 cps, a 2.2-pound "ultimate portable computer" would be able to perform the equivalent of all human thought over the last ten thousand years (assumed at ten billion human brains for ten thousand years) in ten microseconds.⁶⁴ If we examine the "Exponential Growth of Computing" chart (p. 70), we see that this amount of computing is estimated to be available for one thousand dollars by 2080.

Highlight (Yellow) | Location 2339

the 1026 cps represented by biological human civilization today is poorly utilized.

Highlight (Yellow) | Location 2340

2030s will not represent the Singularity, however, because it does not yet correspond to a profound expansion of our intelligence. By the mid-2040s, however, that one thousand dollars' worth of computation will be equal to 1026 cps, so the intelligence created per year (at a total cost of about \$1012) will be about one billion times more powerful than all human intelligence today.⁶⁶

Highlight (Yellow) | Location 2348

Despite the clear predominance of nonbiological intelligence by the mid-2040s, ours will still be a human civilization. We will transcend biology, but not our humanity.

Highlight (Yellow) | Location 2351

By the second decade of this century, however, most computing will not be organized in such rectangular devices but will be highly distributed throughout the environment. Computing will be everywhere: in the walls, in our furniture, in our clothing, and in our bodies and brains.

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higher-resolution technology has been created by scientists at the University of California at Santa Barbara, who have developed an extremely sensitive measurement detector with a physical beam made of gallium-arsenide crystal and a sensing system that can measure a flexing of the beam of as little as one picometer. The device is intended to provide a test of Heisenberg's uncertainty principle.⁷³

Highlight (Yellow) | Location 2400

time dimension Cornell University scientists have demonstrated an imaging technology based on X-ray scattering that can record movies of the movement of a single electron. Each frame represents only four attoseconds (10⁻¹⁸ seconds, each one a billionth of a billionth of a second).⁷⁴ The device can achieve spatial resolution of one angstrom (10⁻¹⁰ meter, which is 100 picometers).

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The mental experiment we should be making is not whether humans as we know them today will be capable of engineering pico- and femtocomputing technologies, but whether the vast intelligence of future nanotechnology-based intelligence (which will be trillions of trillions of times more capable than contemporary biological human intelligence) will be capable of rendering these designs.

Highlight (Yellow) | Location 2417

In addition to making computing smaller, we can make it bigger—that is, we can replicate these very small devices on a massive scale. With full-scale nanotechnology, computing resources can be made self-replicating and thus can rapidly convert mass and energy into an intelligent form.

Highlight (Yellow) | Location 2420

As we will discuss later, there are at least suggestions that the speed of light may not be immutable. Physicists Steve Lamoreaux and Justin Torgerson of the Los Alamos National Laboratory have analyzed data from an old natural nuclear reactor that two billion years ago produced a fission reaction lasting several hundred thousand years in what is now West Africa.⁷⁵ Examining radioactive isotopes left over from the reactor and comparing them to isotopes from similar nuclear reactions today, they determined that the physics constant alpha (also called the fine-structure constant), which determines the strength of the electromagnetic force, apparently has changed over two billion years. This is of great significance to the world of physics, because the speed of light is inversely proportional to alpha, and both have been considered unchangeable constants. Alpha appears to have decreased by 4.5 parts out of 108. If confirmed, this would imply that the speed of light has increased.

Highlight (Yellow) | Location 2433

The speed of light is one of the limits that constrain computing devices even today, so the ability to boost it would extend further the limits of computation.

Highlight (Yellow) | Location 2437

Going Back in Time. Another intriguing—and highly speculative—possibility is to send a computational process back in time through a “wormhole” in space-time. Theoretical physicist Todd Brun of the Institute for Advanced Studies at Princeton

Highlight (Yellow) | Location 2441

Brun does not provide a design for such a device but establishes that such a system is consistent with the laws of physics. His time-traveling computer also does not create the “grandfather paradox,”

Highlight (Yellow) | Location 2520

Reverse Engineering the Brain: An Overview of the Task The combination of human-level intelligence with a computer’s inherent superiority in speed, accuracy, and memory-sharing ability will be formidable.

Highlight (Yellow) | Location 2529

We will also gain powerful new ways to treat neurological problems such as Alzheimer’s, stroke, Parkinson’s disease, and sensory disabilities, and ultimately will be able to vastly extend our intelligence.

Highlight (Yellow) | Location 2553

Once a computer achieves a human level of intelligence, it will necessarily soar past it.

Highlight (Yellow) | Location 2598

As mentioned earlier, auditory-system researcher Lloyd Watts has developed a comprehensive model of a significant portion of the human auditory-processing system from neurobiology studies of specific neuron types and interneuronal-connection information. Watts’s model includes five parallel paths and the actual representations of auditory information at each stage of neural processing. Watts has implemented his model in a computer as real-time software that can locate and identify sounds and functions, similar to the way human hearing operates. Although a work in progress,

Highlight (Yellow) | Location 2621

(Current computers perform computations in billionths of a second, compared to thousandths of a second for interneuronal transactions.)

Highlight (Yellow) | Location 2889

two proteins called zonulin and zot have been discovered that react with receptors in the brain to temporarily open the BBB at select sites. These two proteins play a similar role in opening receptors in the small intestine to allow digestion of glucose and other nutrients.

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the raw signals from the body go through multiple levels of processing before being aggregated in a compact dynamic representation in two small organs called the right and left insula, located deep in the cerebral cortex. For full-immersion virtual reality, it may be more effective to tap into the already-interpreted signals in the insula rather than the unprocessed signals throughout the body.

Highlight (Yellow) | Location 2972

Although chemistry is theoretically based on physics and could be derived entirely from physics, this would be unwieldy and infeasible in practice. So chemistry uses its own rules and models. We should likewise, in theory, be able to deduce the laws of thermodynamics from physics, but this is a far-from-straightforward process. Once we have a sufficient number of particles to call something a gas rather than a bunch of particles, solving equations for each particle interaction becomes impractical, whereas the laws of thermodynamics work extremely well. The interactions of a single molecule within the gas are hopelessly complex and unpredictable, but the gas itself, comprising trillions of molecules, has many predictable properties.

Highlight (Yellow) | Location 2977

It is often unnecessary to express higher-level results using the intricacies of the dynamics of the lower-level systems, although one has to thoroughly understand the lower level before moving to the higher one.

Highlight (Yellow) | Location 3057

“Say a 10-year-old kid uses 1,000 connections to store a piece of information. When he is 80, one-quarter of the connections will still be there, no matter how things change. That’s why you can still remember your childhood experiences.” Gan

Highlight (Yellow) | Location 3062

The reason memories can remain intact even if three quarters of the connections have disappeared is that the coding method used appears to have properties similar to those of a hologram. In a hologram, information is stored in a diffuse pattern throughout an extensive region. If you destroy three quarters of the hologram, the entire image remains intact, although with only one quarter of the resolution. Research by Pentti Kanerva, a neuroscientist at Redwood Neuroscience Institute, supports the idea that memories are dynamically distributed throughout a region of neurons. This explains why older memories persist but nonetheless appear to “fade,” because their resolution has diminished.

Highlight (Yellow) | Location 3186

how does a ten-year-old manage to catch a fly ball?⁷⁴ All that a child can see is the ball's trajectory from his position in the outfield. To actually infer the path of the ball in three-dimensional space would require solving difficult simultaneous differential equations. Additional equations would need to be solved to predict the future course of the ball, and more equations to translate these results into what was required of the player's own movements. How does a young outfielder accomplish all of this in a few seconds with no computer and no training in differential equations?

Highlight (Yellow) | Location 3240

stochastic (random within restrictions) method used to wire the actual human brain from its genetic code.⁸⁵

Highlight (Yellow) | Location 3304

feedforward (lacking feedback) system

Highlight (Yellow) | Location 3376

"A single cubic millimeter of cerebral cortex may contain on the order of 5 billion ... synapses of different shapes and sizes." The cortex is responsible for perception, planning, decision making and most of what we regard as conscious thinking. Our ability to use language, another unique attribute of our species, appears to be located in this region.

Highlight (Yellow) | Location 3381

Theorists Giacomo Rizzolatti and Michael Arbib hypothesized that language emerged from manual gestures (which monkeys—and, of course, humans—are capable of). Performing manual gestures requires the ability to mentally correlate the performance and observation of one's own hand movements.¹¹¹ Their "mirror system hypothesis" is that the key to the evolution of language is a property called "parity," which is the understanding that the gesture (or utterance) has the same meaning for the party making the gesture as for the party receiving it; that is, the understanding that what you see in a mirror is the same (although reversed left-to-right) as what is seen by someone else watching you. Other animals are unable to understand the image in a mirror in this fashion, and it is believed that they are missing this key ability to deploy parity. A closely related concept is that the ability to imitate the movements (or, in the case of human babies, vocal sounds) of others is critical to developing language.¹¹² Imitation requires the ability to break down an observed presentation into parts, each of which can then be mastered through recursive and iterative refinement. Recursion is the key capability identified in a new theory of linguistic competence. In Noam Chomsky's early theories of language in humans, he cited many common attributes that account for the similarities in human languages. In a 2002 paper by Marc Hauser, Noam Chomsky, and Tecumseh Fitch, the authors cite the single attribution of "recursion" as accounting for the unique language faculty of the human species.¹¹³ Recursion is the ability to put together small parts into a larger chunk, and then use that chunk as a part in yet another structure and to continue this process iteratively.

Highlight (Yellow) | Location 3398

Interestingly, we are able to predict or anticipate our own decisions. Work by physiology professor Benjamin Libet at the University of California at Davis shows that neural activity to initiate an action actually occurs about a third of a second before the brain has made the decision to take the action. The implication, according to Libet, is that the decision is really an illusion, that “consciousness is out of the loop.” The cognitive scientist and philosopher Daniel Dennett describes the phenomenon as follows: “The action is originally precipitated in some part of the brain, and off fly the signals to muscles, pausing en route to tell you, the conscious agent, what is going on (but like all good officials letting you, the bumbling president, maintain the illusion that you started it all).”¹¹⁴

Highlight (Yellow) | Location 3405

in patients whose left and right brains are no longer connected, one side of the brain (usually the more verbal left side) will create elaborate explanations (“confabulations”) for actions initiated by the other side, as if the left side were the public-relations agent for the right side.

Highlight (Yellow) | Location 3415

emotionally charged situations appear to be handled by special cells called spindle cells, which are found only in humans and some great apes. These neural cells are large, with long neural filaments called apical dendrites that connect extensive signals from many other brain regions. This type of “deep” interconnectedness,

Highlight (Yellow) | Location 3420

startling, however, is how few spindle cells there are in this tiny region: only about 80,000 in the human brain (about 45,000 in the right hemisphere and 35,000 in the left hemisphere). This disparity appears to account for the perception that emotional intelligence is the province of the right brain, although the disproportion is modest. Gorillas have about 16,000 of these cells, bonobos about 2,100, and chimpanzees about 1,800. Other mammals lack them completely.

Highlight (Yellow) | Location 3436

growing consensus that our emotions are closely linked to areas of the brain that contain maps of the body, a view promoted by Dr. Antonio Damasio at the University of Iowa.¹¹⁶ They are also consistent with the view that a great deal of our thinking is directed toward our bodies: protecting and enhancing them, as well as attending to their myriad needs and desires.

Highlight (Yellow) | Location 3445

spindle cells do not exist in newborn humans but begin to appear only at around the age of four months and increase significantly from ages one to three. Children’s ability to deal with moral issues and perceive such higher-level emotions as love develop during this same time period. The spindle cells gain their power from the deep interconnectedness of their long apical dendrites with many other brain regions. The high-level emotions that the spindle cells process are affected, thereby, by all of our perceptual and cognitive regions.

Highlight (Yellow) | Location 3450

We have fifty billion neurons in the cerebellum that deal with skill formation, billions in the cortex that perform the transformations for perception and rational planning, but only about eighty thousand spindle cells dealing with high-level emotions. It is important to point out that the spindle cells are not doing rational problem solving, which is why we don't have rational control over our responses to music or over falling in love. The rest of the brain is heavily engaged, however, in trying to make sense of our mysterious high-level emotions. Interfacing the Brain and Machines I want to do something with my life; I want to be a cyborg. —KEVIN WARWICK

Highlight (Yellow) | Location 3463

Miguel Nicolelis and his colleagues at Duke University implanted sensors in the brains of monkeys, enabling the animals to control a robot through thought alone. The first step in the experiment involved teaching the monkeys to control a cursor on a screen with a joystick. The scientists collected a pattern of signals from EEGs (brain sensors) and subsequently caused the cursor to respond to the appropriate patterns rather than physical movements of the joystick. The monkeys quickly learned that the joystick was no longer operative and that they could control the cursor just by thinking. This "thought detection" system was then hooked up to a robot, and the monkeys were able to learn how to control the robot's movements with their thoughts alone. By getting visual feedback on the robot's performance, the monkeys were able to perfect their thought control over the robot. The goal of this research is to provide a similar system for paralyzed humans that will enable them to control their limbs and environment.

Highlight (Yellow) | Location 3472

Another approach being pursued by the Max Planck Institute for Human Cognitive and Brain Sciences in Munich is directly interfacing nerves and electronic devices. A chip created by Infineon allows neurons to grow on a special substrate that provides direct contact between nerves and electronic sensors and stimulators. Similar work on a "neurochip" at Caltech has demonstrated two-way, noninvasive communication between neurons and electronics.¹¹⁷ We have already learned how to interface surgically installed neural implants. In cochlear (inner-ear) implants it has been found that the auditory nerve reorganizes itself to correctly interpret the multichannel signal from the implant.

Highlight (Yellow) | Location 3486

We know what we are, but know not what we may be. —WILLIAM SHAKESPEARE

Highlight (Yellow) | Location 3518

Once the nanobot era arrives in the 2020s we will be able to observe all of the relevant features of neural performance with very high resolution from inside the brain itself. Sending billions of nanobots through its capillaries will enable us to noninvasively scan an entire working brain in real time.

Highlight (Yellow) | Location 3523

which will make it possible to combine the principles of operation of human intelligence with the forms of intelligent information processing that we have derived from other AI research. We will also benefit from the inherent strength of machines in storing, retrieving, and quickly sharing massive amounts of information. We will then be in a position to implement these powerful hybrid systems on computational platforms that greatly exceed the capabilities of the human brain's relatively fixed architecture.

Highlight (Yellow) | Location 3538

That our intelligence is just above the threshold necessary to understand itself results from our native ability, combined with the tools of our own making, to envision, refine, extend, and alter abstract—and increasingly subtle—models of our own observations.

Highlight (Yellow) | Location 3547

If we are truly capturing a particular person's mental processes, then the reinstantiated mind will need a body, since so much of our thinking is directed toward physical needs and desires. As I will discuss in chapter 5, by the time we have the tools to capture and re-create a human brain with all of its subtleties, we will have plenty of options for twenty-first-century bodies for both nonbiological humans and biological humans who avail themselves of extensions to our intelligence. The human body version 2.0 will include virtual bodies in completely realistic virtual environments,

Highlight (Yellow) | Location 3553

Although I estimated that 1016 cps of computation and 1013 bits of memory are sufficient to emulate human levels of intelligence, my estimates for the requirements of uploading were higher: 1019 cps and 1018 bits, respectively.

Highlight (Yellow) | Location 3560

The basic computational resources (1019 cps and 1018 bits) will be available for one thousand dollars in the early 2030s, about a decade later than the resources needed for functional simulation.

Highlight (Yellow) | Location 3568

That confirmation will be in the form of a functional simulation of human intelligence that passes the Turing test, which I believe will take place by 2029.¹¹⁹ To capture this level of detail will require scanning from within the brain using nanobots, the technology for which will be available by the late 2020s. Thus, the early 2030s is a reasonable time frame for the computational performance, memory, and brain-scanning prerequisites of uploading. Like any other technology, it will take some iterative refinement to perfect this capability, so the end of the 2030s is a conservative projection for successful uploading.

Highlight (Yellow) | Location 3578

By that time we'll face some complications in devising the rules of any Turing test. Since nonbiological intelligence will have passed the original Turing test years earlier (around 2029), should we allow a nonbiological human equivalent to be a judge? How about an enhanced human? Unenhanced humans may become increasingly hard to find.

Highlight (Yellow) | Location 3607

In the 2020s we will use nanobots to begin augmenting our brains with nonbiological intelligence, starting with the "routine" functions of sensory processing and memory, moving on to skill formation, pattern recognition, and logical analysis. By the 2030s the nonbiological portion of our intelligence will predominate, and by the 2040s, as I pointed out in chapter 3, the nonbiological portion will be billions of times more capable. Although we are likely to retain the biological portion for a period of time, it will become of increasingly little consequence. So we will have effectively uploaded ourselves, albeit gradually, never quite noticing the transfer.

Highlight (Yellow) | Location 3613

Although I believe that uploading as in the sudden scan-and-transfer scenario discussed in this section will be a feature of our future world, it is this gradual but inexorable progression to vastly superior nonbiological thinking that will profoundly transform human civilization.

Highlight (Yellow) | Location 3619

Given the human brain's plasticity, our thoughts literally create our brains through the growth of new spines, synapses, dendrites, and even neurons. As a result, Einstein's parietal lobes—the region associated with visual imagery and mathematical thinking—became greatly enlarged.¹²¹

Highlight (Yellow) | Location 3632

when one AI learns something, it can quickly share that knowledge with many other AIs. RAY: They'll have access to all of our exponentially growing knowledge on the Web, which will include habitable, full-immersion virtual-reality environments where they can interact with one another and with biological humans who are projecting themselves into these environments.

Highlight (Yellow) | Location 3655

Keep in mind, there won't be a clear distinction between AIs and humans. MOLLY 2104: Yes, except for the MOSHs (Mostly Original Substrate Humans) of course.

Highlight (Yellow) | Location 3673

The first half of the twenty-first century will be characterized by three overlapping revolutions—in Genetics, Nanotechnology, and Robotics. These will usher in what I referred to earlier as Epoch Five, the beginning of the Singularity.

Highlight (Yellow) | Location 3848

emerging field of rejuvenation medicine. For example, we will be able to create new heart cells from skin cells and introduce them into the system through the bloodstream. Over time, existing heart cells will be replaced with these new cells, and the result will be a rejuvenated “young” heart manufactured using a person’s own DNA. I discuss this approach to regrowing our bodies below.

Highlight (Yellow) | Location 3878

Viruses are often the vehicle of choice. Long ago viruses learned how to deliver their genetic material to human cells and, as a result, cause disease. Researchers now simply switch the material a virus unloads into cells by removing its genes and inserting therapeutic ones.

Highlight (Yellow) | Location 3890

Even faced with these obstacles gene therapy is starting to work in human applications. A team led by University of Glasgow research doctor Andrew H. Baker has successfully used adenoviruses to “infect” specific organs and even specific regions within organs. For example, the group was able to direct gene therapy precisely at the endothelial cells, which line the inside of blood vessels. Another approach is being developed by Celera Genomics, a company founded by Craig Venter (the head of the private effort to transcribe the human genome). Celera has already demonstrated the ability to create synthetic viruses from genetic information and plans to apply these biodesigned viruses to gene therapy.³⁷

Highlight (Yellow) | Location 3901

Reversing Degenerative Disease Degenerative (progressive) diseases—heart disease, stroke, cancer, type 2 diabetes, liver disease, and kidney disease—account for about 90 percent of the deaths in our society. Our understanding of the principal components of degenerative disease and human aging is growing rapidly, and strategies have been identified to halt and even reverse each of these processes. In *Fantastic Voyage*, Grossman and I describe a wide range of therapies now in the testing pipeline that have already demonstrated significant results in attacking the key biochemical steps underlying the progress of such diseases.

Highlight (Yellow) | Location 4015

Preserving Endangered Species and Restoring Extinct Ones. Another exciting application is re-creating animals from endangered species. By cryopreserving cells from these species, they never need become extinct. It will eventually be possible to re-create animals from recently extinct species. In 2001 scientists were able to synthesize DNA for the Tasmanian tiger, which had then been extinct for sixty-five years, with the hope of bringing this species back to life.⁶⁰ As for long-extinct species (for example, dinosaurs), it is highly doubtful that we will find the fully intact DNA required in a single preserved cell (as they did in the movie *Jurassic Park*). It is likely, however, that we will eventually be able to synthesize the necessary DNA by patching together the information derived from multiple inactive fragments.

Highlight (Yellow) | Location 4035

to avoid autoimmune reactions, you would strongly prefer to obtain these with your own DNA rather than the DNA from someone else's germ-line cells. In addition, this approach uses plentiful skin cells (of the patient) rather than rare and precious stem cells. Transdifferentiation will directly grow an organ with your genetic makeup. Perhaps most important, the new organ can have its telomeres fully extended to their original youthful length, so that the new organ is effectively young again.⁶⁵

Highlight (Yellow) | Location 4041

Current treatments for type 1 diabetes require strong antirejection drugs that can have dangerous side effects.⁶⁶ With somatic-cell engineering, type 1 diabetics will be able to make pancreatic islet cells from their own cells, either from skin cells (transdifferentiation) or from adult stem cells. They would be using their own DNA, and drawing upon a relatively inexhaustible supply of cells, so no antirejection drugs would be required. (But to fully cure type 1 diabetes, we would also have to overcome the patient's autoimmune disorder, which causes his body to destroy islet cells.)

Highlight (Yellow) | Location 4045

Even more exciting is the prospect of replacing one's organs and tissues with their "young" replacements without surgery. Introducing cloned, telomere-extended, DNA-corrected cells into an organ will allow them to integrate themselves with the older cells. By repeated treatments of this kind over a period of time, the organ will end up being dominated by the younger cells. We normally replace our own cells on a regular basis anyway, so why not do so with youthful rejuvenated cells rather than telomere-shortened error-filled ones? There's no reason why we couldn't repeat this process for every organ and tissue in our body, enabling us to grow progressively younger. Solving World Hunger. Cloning technologies even offer a possible solution for world hunger: creating meat and other protein sources in a factory without animals by cloning animal muscle tissue. Benefits would include extremely low cost, avoidance of pesticides and hormones that occur in natural meat, greatly reduced environmental impact (compared to factory farming), improved nutritional profile, and no animal suffering. As with therapeutic cloning, we would not be creating the entire animal but rather directly producing the desired animal parts or flesh. Essentially, all of the meat—billions of pounds of it—would be derived from a single animal.

Highlight (Orange) | Location 4059

The advent of animal-less meat will also eliminate animal suffering. The economics of factory farming place a very low priority on the comfort of animals, which are treated as cogs in a machine. The meat produced in this manner, although normal in all other respects, would not be part of an animal with a nervous system, which is generally regarded as a necessary element for suffering

Highlight (Yellow) | Location 4067

Physical cloning is far different from mental cloning, in which a person's entire personality, memory, skills, and history will ultimately be downloaded into a different, and most likely more powerful, thinking medium. There's

no issue of philosophical identity with genetic cloning, since such clones would be different people, even more so than conventional twins are today.

Highlight (Yellow) | Location 4112

We are shrinking the key feature size of technology, in accordance with the law of accelerating returns, at the exponential rate of approximately a factor of four per linear dimension per decade.⁶⁸ At this rate the key feature sizes for most electronic and many mechanical technologies will be in the nanotechnology range—generally considered to be under one hundred nanometers—by the 2020s.

Highlight (Yellow) | Location 4122

The revolution in nanotechnology, however, will ultimately enable us to redesign and rebuild, molecule by molecule, our bodies and brains and the world with which we interact.⁷⁰ These two revolutions are overlapping, but the full realization of nanotechnology lags behind the biotechnology revolution by about one decade. Most nanotechnology historians date the conceptual birth of nanotechnology to physicist Richard Feynman’s seminal speech in 1959, “There’s Plenty of Room at the Bottom,” in which he described the inevitability and profound implications of engineering machines at the level of atoms:

Highlight (Yellow) | Location 4132

An even earlier conceptual foundation for nanotechnology was formulated by the information theorist John von Neumann in the early 1950s with his model of a self-replicating system based on a universal constructor, combined with a universal computer.⁷²

Highlight (Yellow) | Location 4137

he took the concept one step further and proposed a “kinematic constructor”: a robot with at least one manipulator (arm) that would build a replica of itself from a “sea of parts” in its midst.⁷³

Highlight (Yellow) | Location 4142

Drexler’s dissertation (which became his book *Engines of Creation* in 1986 and was articulated technically in his 1992 book, *Nanosystems*) laid out the foundation of nanotechnology and provided the road map still being followed today.⁷⁴ Drexler’s “molecular assembler” will be able to make almost anything in the world. It has been referred to as a “universal assembler,”

Highlight (Yellow) | Location 4191

Although many configurations have been proposed, the typical assembler has been described as a tabletop unit that can manufacture almost any physically possible product for which we have a software description, ranging from computers, clothes, and works of art to cooked meals.⁷⁹ Larger products, such as furniture, cars, or even houses, can be built in a modular fashion or using larger assemblers. Of particular importance is the fact that an assembler can create copies of itself, unless its design specifically prohibits this (to avoid potentially dangerous

self-replication). The incremental cost of creating any physical product, including the assemblers themselves, would be pennies per pound—basically the cost of the raw materials. Drexler estimates total manufacturing cost for a molecular-manufacturing process in the range of ten cents to fifty cents per kilogram, regardless of whether the manufactured product were clothing, massively parallel supercomputers, or additional manufacturing systems.⁸⁰ The real cost, of course, would be the value of the information describing each type of product—that is, the software that controls the assembly process. In other words, the value of everything in the world, including physical objects, would be based essentially on information.

Highlight (Yellow) | Location 4206

In operation, the centralized data store would send out commands simultaneously to many trillions (some estimates as high as 1018) of robots in an assembler, each receiving the same instruction at the same time.

Highlight (Yellow) | Location 4215

The Biological Assembler Nature shows that molecules can serve as machines because living things work by means of such machinery. Enzymes are molecular machines that make, break, and rearrange the bonds holding other molecules together. Muscles are driven by molecular machines that haul fibers past one another. DNA serves as a data-storage system, transmitting digital instructions to molecular machines, the ribosomes, that manufacture protein molecules. And these protein molecules, in turn, make up most of the molecular machinery. —ERIC DREXLER The ultimate existence proof of the feasibility of a molecular assembler is life itself.

Highlight (Yellow) | Location 4254

one of several protections against molecular-scale replication of robots in an uncontrolled fashion in the outside world. Biology's replication robot, the ribosome, also requires carefully controlled source and fuel materials, which are provided by our digestive system.

Highlight (Yellow) | Location 4261

construction of related molecular machines. Boston College chemistry professor T. Ross Kelly reported that he constructed a chemically powered nanomotor out of seventy-eight atoms.⁸² A biomolecular research group headed by Carlo Montemagno created an ATP-fueled nanomotor.⁸³ Another molecule-sized motor fueled by solar energy was created out of fifty-eight atoms by Ben Feringa at the University of Groningen in the Netherlands.⁸⁴

Highlight (Yellow) | Location 4276

A study conducted for NASA by General Dynamics has demonstrated the feasibility of self-replicating nanoscale machines.⁸⁶

Highlight (Yellow) | Location 4292

A particularly impressive demonstration of a nanoscale device constructed from DNA is a tiny biped robot that can walk on legs that are ten nanometers long.⁹⁰ Both the legs and the walking track are built from DNA, again

chosen for the molecule's ability to attach and detach itself in a controlled manner. The nanorobot, a project of chemistry professors Nadrian Seeman and William Sherman of New York University, walks by detaching its legs from the track, moving down it, and then reattaching its legs to the track. The project is another impressive demonstration of the ability of nanoscale machines to execute precise maneuvers.

Highlight (Yellow) | Location 4301

Another approach is to use only parts of bacteria. A research group headed by Viola Vogel at the University of Washington built a system using just the limbs of *E. coli* bacteria that was able to sort out nanoscale beads of different sizes. Since bacteria are natural nanoscale systems that can perform a wide variety of functions, the ultimate goal of this research will be to reverse engineer the bacteria so that the same design principles can be applied to our own nanobot designs.

Highlight (Yellow) | Location 4327

Smalley is correct, of course, that an assembler with only one nanobot wouldn't produce any appreciable quantities of a product. However, the basic concept of nanotechnology is that we will use trillions of nanobots to accomplish meaningful results—a factor that is also the source of the safety concerns that have received so much attention. Creating this many nanobots at reasonable cost will require self-replication at some level, which while solving the economic issue will introduce potentially grave dangers, a concern I will address in chapter 8. Biology uses the same solution to create organisms with trillions of cells, and indeed we find that virtually all diseases derive from biology's self-replication process gone awry.

Highlight (Yellow) | Location 4335

conceptual designers of nanotechnology have emphasized building structural components from diamondoid or carbon nanotubes. Increasing the strength or stiffness of a system reduces its susceptibility to thermal effects. Analysis of these designs has shown them to be thousands of times more stable in the presence of thermal effects than are biological systems, so they can operate in a far wider temperature range.⁹⁶

Highlight (Yellow) | Location 4351

Smalley's own observation, "When a scientist says something is possible, they're probably underestimating how long it will take. But if they say it's impossible, they're probably wrong."

Highlight (Yellow) | Location 4367

the primary thrust of our technology has been to develop systems that are not limited to the restrictions of biological evolution, which exclusively adopted water-based chemistry and proteins as its foundation. Biological systems can fly, but if you want to fly at thirty thousand feet and at hundreds or thousands of miles per hour, you would use our modern technology, not proteins. Biological systems such as human brains can remember things and do calculations, but if you want to do data mining on billions of items of information, you would want to use electronic technology, not unassisted human brains.

Highlight (Yellow) | Location 4413

By the 2020s molecular assembly will provide tools to effectively combat poverty, clean up our environment, overcome disease, extend human longevity, and many other worthwhile pursuits. Like every other technology that humankind has created, it can also be used to amplify and enable our destructive side. It's important that we approach this technology in a knowledgeable manner to gain the profound benefits it promises, while avoiding its dangers.

Highlight (Yellow) | Location 4435

MicroCHIPS of Bedford, Massachusetts, has developed a computerized device that is implanted under the skin and delivers precise mixtures of medicines from hundreds of nanoscale wells inside the device.¹¹³ Future versions of the device are expected to be able to measure blood levels of substances such as glucose. The system could be used as an artificial pancreas, releasing precise amounts of insulin based on blood glucose response. It would also be capable of simulating any other hormone-producing organ. If trials go smoothly, the system could be on the market by 2008.

Highlight (Yellow) | Location 4441

Nanoscale packages can be designed to contain drugs, protect them through the GI tract, guide them to specific locations, and then release them in sophisticated ways, including allowing them to receive instructions from outside the body. Nanotherapeutics in Alachua, Florida, has developed a biodegradable polymer only several nanometers thick that uses this approach.¹¹⁴

Highlight (Yellow) | Location 4444

Powering the Singularity We produce about 14 trillion (about 10¹³) watts of power today in the world. Of this energy about 33 percent comes from oil, 25 percent from coal, 20 percent from gas, 7 percent from nuclear fission reactors, 15 percent from biomass and hydroelectric sources, and only 0.5 percent from renewable solar, wind, and geothermal technologies.¹¹⁵ Most air pollution and significant contributions to water and other forms of pollution result from the extraction, transportation, processing, and uses of the 78 percent of our energy that comes from fossil fuels. The energy obtained from oil also contributes to geopolitical tensions, and there's the small matter of its \$2 trillion per year price tag for all of this energy.

Highlight (Yellow) | Location 4454

A primary implication of the nanotechnology revolution is that physical technologies, such as manufacturing and energy, will become governed by the law of accelerating returns. All technologies will essentially become information technologies, including energy.

Highlight (Yellow) | Location 4457

The bulk of the additional energy needed is likely to come from new nanoscale solar, wind, and geothermal technologies. It's important to recognize that most energy sources today represent solar power in one form or another.

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The energy requirements for nanofactories are negligible. Drexler estimates that molecular manufacturing will be an energy generator rather than an energy consumer. According to Drexler, “A molecular manufacturing process can be driven by the chemical energy content of the feedstock materials, producing electrical energy as a by-product (if only to reduce the heat dissipation burden)... Using typical organic feedstock, and assuming oxidation of surplus hydrogen, reasonably efficient molecular manufacturing processes are net energy producers.”¹¹⁸

Highlight (Yellow) | Location 4491

Nanotechnology-based lighting will use small, cool, light-emitting diodes, quantum dots, or other innovative light sources to replace hot, inefficient incandescent and fluorescent bulbs.

Highlight (Yellow) | Location 4519

Another option for energy transmission is wireless transmission by microwaves. This method would be especially well suited to efficiently beam energy created in space by giant solar panels (see below).¹²¹ The Millennium Project of the American Council for the United Nations University envisions microwave energy transmission as a key aspect of “a clean, abundant energy future.”¹²²

Highlight (Yellow) | Location 4543

Scientists at the University of Texas have developed a nanobot-size fuel cell that produces electricity directly from the glucose-oxygen reaction in human blood.¹²⁸ Called a “vampire bot” by commentators,

Highlight (Yellow) | Location 4546

system had the theoretical potential to produce a peak of one hundred watts from the blood of one person, although implantable devices would use far less. (A newspaper in Sydney observed that the project provided a basis for the premise in the Matrix movies of using humans as batteries.)¹²⁹ Another approach to converting the abundant sugar found in the natural world into electricity has been demonstrated by Swades K. Chaudhuri and Derek R. Lovley at the University of Massachusetts. Their fuel cell, which incorporates actual microbes (the *Rhodospirillum rubrum* bacterium), boasts a remarkable 81 percent efficiency and uses almost no energy in its idling mode. The bacteria produce electricity directly from glucose with no unstable intermediary by-products. The bacteria also use the sugar fuel to reproduce, thereby replenishing themselves, resulting in stable and continuous production of electrical energy.

Highlight (Yellow) | Location 4563

despite the enormous increases in computation and communication over the next quarter century and the resulting economic growth, the far greater energy efficiencies of nanotechnology imply that energy requirements will increase only modestly to around thirty trillion watts (3×10^{13}) by 2030. We could meet this entire energy need with solar power alone if we captured only 0.0003 (three ten-thousandths) of the sun’s energy as it hits the

Earth. It's interesting to compare these figures to the total metabolic energy output of all humans, estimated by Robert Freitas at 1012 watts, and that of all vegetation on Earth, at 1014 watts.

Highlight (Yellow) | Location 4568

Freitas also estimates that the amount of energy we could produce and use without disrupting the global energy balance required to maintain current biological ecology (referred to by climatologists as the "hypsithermal limit") is around 1015 watts. This would allow a very substantial number of nanobots per person for intelligence enhancement and medical purposes, as well as other applications, such as providing energy and cleaning up the environment. Estimating a global population of around ten billion (1010) humans, Freitas estimates around 1016 (ten thousand trillion) nanobots for each human would be acceptable within this limit.¹³² We would need only 1011 nanobots (ten millionths of this limit) per person to place one in every neuron. By the time we have technology of this scale, we will also be able to apply nanotechnology to recycle energy by capturing at least a significant portion of the heat generated by nanobots and other nanomachinery and converting that heat back into energy. The most effective way to do this would probably be to build the energy recycling into the nanobot itself.¹³³ This is similar to the idea of reversible logic gates in computation, in which each logic gate essentially immediately recycles the energy it used for its last computation.

Highlight (Yellow) | Location 4577

We could also pull carbon dioxide out of the atmosphere to provide the carbon for nanomachinery, which would reverse the increase in carbon dioxide resulting from our current industrial-era technologies. We might, however, want to be particularly cautious about doing more than reversing the increase over the past several decades, lest we replace global warming with global cooling.

Highlight (Yellow) | Location 4586

Industry sources indicate that once solar power falls below \$1.00 per watt, it will be competitive for directly supplying electricity to the nation's power grid. Nanosolar has a design based on titanium oxide nanoparticles that can be mass-produced on very thin flexible films. CEO Martin Roscheisen estimates that his technology has the potential to bring down solar-power costs to around fifty cents per watt by 2006, lower than that of natural gas.¹³⁷ Competitors Nanosys and Konarka have similar projections. Whether or not these business plans pan out, once we have MNT (molecular nanotechnology)-based manufacturing, we will be able to produce solar panels (and almost everything else) extremely inexpensively, essentially at the cost of raw materials, of which inexpensive carbon is the primary one. At an estimated thickness of several microns, solar panels could ultimately be as inexpensive as a penny per square meter.

Highlight (Yellow) | Location 4594

Terrestrial surfaces could be augmented by huge solar panels in space. A Space Solar Power satellite already designed by NASA could convert sunlight in space to electricity and beam it to Earth by microwave. Each such satellite could provide billions of watts of electricity, enough for tens of thousands of homes.¹³⁸ With circa-2029 MNT manufacturing, we could produce solar panels of vast size directly in orbit around the Earth, requiring only the shipment of the raw materials to space stations, possibly via the planned Space Elevator, a

thin ribbon, extending from a shipborne anchor to a counterweight well beyond geosynchronous orbit, made out of a material called carbon nanotube composite.¹³⁹

Highlight (Yellow) | Location 4620

As with all new technologies, there is a downside to nanoparticles: the introduction of new forms of toxins and other unanticipated interactions with the environment and life. Many toxic materials, such as gallium arsenide, are already entering the ecosystem through discarded electronic products. The same properties that enable nanoparticles and nanolayers to deliver highly targeted beneficial results can also lead to unforeseen reactions, particularly with biological systems such as our food supply and our own bodies.

Highlight (Yellow) | Location 4632

Nanofiltration membranes for water purification provide dramatically improved removal of fine-particle contaminants, compared to conventional methods of using sedimentation basins and wastewater clarifiers. Nanoparticles with designed catalysis are capable of absorbing and removing impurities. By using magnetic separation, these nanomaterials can be reused, which prevents them from becoming contaminants themselves.

Highlight (Yellow) | Location 4641

It's estimated that the widespread use of nanocomposites for structural material in automobiles would reduce gasoline consumption by 1.5 billion liters

Highlight (Yellow) | Location 4649

New computer displays using nanotube-based field-emission displays (FEDs) will provide superior display specifications while eliminating the heavy metals and other toxic materials used in conventional displays.

Highlight (Yellow) | Location 4655

Once we can go beyond simple nanoparticles and nanolayers and create more complex systems through precisely controlled molecular nanoassembly, we will be in a position to create massive numbers of tiny intelligent devices capable of carrying out relatively complex tasks. Cleaning up the environment will certainly be one of those missions.

Highlight (Yellow) | Location 4661

The net effect of these nanomedical interventions will be the continuing arrest of all biological aging, along with the reduction of current biological age to whatever new biological age is deemed desirable by the patient, severing forever the link between calendar time and biological health. Such interventions may become commonplace several decades from today. Using annual checkups and cleanouts, and some occasional major repairs, your biological age could be restored once a year to the more or less constant physiological age that you select. You might still eventually die of accidental causes, but you'll live at least ten times longer than you do now. —ROBERT A. FREITAS JR.¹⁴⁸

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deployment of billions or trillions of nanobots: small robots the size of human blood cells or smaller that can travel inside the bloodstream. This notion is not as futuristic as it may sound; successful animal experiments have been conducted using this concept, and many such microscale devices are already working in animals. At least four major conferences on BioMEMS (Biological Micro Electronic Mechanical Systems) deal with devices to be used in the human bloodstream.¹⁴⁹

Highlight (Yellow) | Location 4673

based on miniaturization and cost-reduction trends, will be feasible within about twenty-five years.

Highlight (Yellow) | Location 4675

Robert A. Freitas Jr.—a pioneering nanotechnology theorist and leading proponent of nanomedicine (reconfiguring our biological systems through engineering on a molecular scale), and author of a book with that title¹⁵⁰—has designed robotic replacements for human blood cells that perform hundreds or thousands of times more effectively than their biological counterparts. With Freitas’s respirocytes (robotic red blood cells) a runner could do an Olympic sprint for fifteen minutes without taking a breath.¹⁵¹

Highlight (Yellow) | Location 4694

a researcher at the University of Illinois at Chicago has cured type 1 diabetes in rats with a nanoengineered device that incorporates pancreatic islet cells.¹⁵⁶

Highlight (Yellow) | Location 4697

MOLLY 2004: Okay, so I’ll have all these nanobots in my bloodstream. Aside from being able to sit at the bottom of my pool for hours, what else is this going to do for me? RAY: It will keep you healthy. They’ll destroy pathogens such as bacteria, viruses, and cancer cells, and they won’t be subject to the various pitfalls of the immune system, such as autoimmune reactions. Unlike your biological immune system, if you don’t like what the nanobots are doing, you can tell them to do something different. MOLLY 2004: You mean, send my nanobots an e-mail? Like, Hey, nanobots, stop destroying those bacteria in my intestines because they’re actually good for my digestion? RAY: Yes, good example. The nanobots will be under our control. They’ll communicate with one another and with the Internet. Even today we have neural implants (for example, for Parkinson’s disease) that allow the patient to download new software into them. MOLLY 2004: That kind of makes the software-virus issue a lot more serious, doesn’t it? Right now, if I get hit with a bad software virus, I may have to run a virus-cleansing program and load my backup files, but if nanobots in my bloodstream get a rogue message, they may start destroying my blood cells. RAY: Well, that’s another reason you’ll probably want robotic blood cells, but your point is well taken. However, it’s not a new issue. Even in 2004, we already have mission-critical software systems that run intensive-care units, manage 911 emergency systems, control nuclear-power plants, land airplanes, and guide cruise missiles. So software integrity is already of critical importance. MOLLY 2004: True, but the idea of software running in my body and brain seems more daunting. On my personal computer, I get more than one hundred spam messages a day, at least several of which contain

malicious software viruses. I'm not real comfortable with nanobots in my body getting software viruses. RAY: You're thinking in terms of conventional Internet access. With VPNs (private networks), we already have the means today to create secure firewalls—otherwise, contemporary mission-critical systems would be impossible. They do work reasonably well, and Internet security technology will continue to evolve. MOLLY 2004: I think some people would take issue with your confidence in firewalls. RAY: They're not perfect, true, and they never will be, but we have another couple decades before we'll have extensive software running in our bodies and brains. MOLLY 2004: Okay, but the virus writers will be improving their craft as well. RAY: It's going to be a nervous standoff, no question about it. But the benefit today clearly outweighs the damage. MOLLY 2004: How clear is that? RAY: Well, no one is seriously arguing we should do away with the Internet because software viruses are such a big problem. MOLLY 2004: I'll give you that. RAY: When nanotechnology is mature, it's going to solve the problems of biology by overcoming biological pathogens, removing toxins, correcting DNA errors, and reversing other sources of aging. We will then have to contend with new dangers that it introduces, just as the Internet introduced the danger of software viruses. These new pitfalls will include the potential for self-replicating nanotechnology getting out of control, as well as the integrity of the software controlling these powerful, distributed nanobots. MOLLY 2004: Did you say reverse aging? RAY: I see you're already picking up on a key benefit. MOLLY 2004: So how are the nanobots going to do that? RAY: We'll actually accomplish most of that with biotechnology, methods such as RNA interference for turning off destructive genes, gene therapy for changing your genetic code, therapeutic cloning for regenerating your cells and tissues, smart drugs to reprogram your metabolic pathways, and many other emerging techniques. But whatever biotechnology doesn't get around to accomplishing, we'll have the means to do with nanotechnology. MOLLY 2004: Such as? RAY: Nanobots will be able to travel through the bloodstream, then go in and around our cells and perform various services, such as removing toxins, sweeping out debris, correcting DNA errors, repairing and restoring cell membranes, reversing atherosclerosis, modifying the levels of hormones, neurotransmitters, and other metabolic chemicals, and a myriad of other tasks. For each aging process, we can describe a means for nanobots to reverse the process, down to the level of individual cells, cell components, and molecules. MOLLY 2004: So I'll stay young indefinitely? RAY: That's the idea. MOLLY 2004: When did you say I could get these? RAY: I thought you were worried about nanobot firewalls. MOLLY 2004: Yeah, well, I've got time to worry about that. So what was that time frame again? RAY: About twenty to twenty-five years.

Highlight (Yellow) | Location 4769

MOLLY 2104: In the 2040s we developed the means to instantly create new portions of ourselves, either biological or nonbiological. It became apparent that our true nature was a pattern of information, but we still needed to manifest ourselves in some physical form. However, we could quickly change that physical form. MOLLY 2004: By? MOLLY 2104: By applying new high-speed MNT manufacturing. So we could readily and rapidly redesign our physical instantiation. So I could have a biological body at one time and not at another, then have it again, then change it, and so on.

Highlight (Yellow) | Location 4779

so why bother with physical instantiations? MOLLY 2004: Yeah, it's messy isn't it? MOLLY 2104: I'll say. MOLLY 2004: I do have to say that it seems strange to be able to change your physical embodiment. I mean, where's your—my—continuity? MOLLY 2104: It's the same as your continuity in 2004. You're changing your particles all the time also. It's just your pattern of information that has continuity. MOLLY 2004: But in 2104

you're able to change your pattern of information quickly also. I can't do that yet. MOLLY 2104: It's really not that different. You change your pattern—your memory, skills, experiences, even personality over time—but there is a continuity, a core that changes only gradually. MOLLY 2004: But I thought you could change your appearance and personality dramatically in an instant? MOLLY 2104: Yes, but that's just a surface manifestation. My true core changes only gradually, just like when I was you

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People want better computers and smarter software, and they want the benefits these machines can help produce. Better medical drugs; relief for humans from the need to perform boring or dangerous jobs; entertainment—there is no end to the list of consumer-benefits. There is also a strong military motive to develop artificial intelligence. And nowhere on the path is there any natural stopping point where technophobics could plausibly argue “hither but not further.” —NICK BOSTROM, “HOW LONG BEFORE SUPERINTELLIGENCE?” 1997

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While the R in GNR stands for robotics, the real issue involved here is strong AI (artificial intelligence that exceeds human intelligence). The standard reason for emphasizing robotics in this formulation is that intelligence needs an embodiment, a physical presence, to affect the world. I disagree with the emphasis on physical presence, however, for I believe that the central concern is intelligence. Intelligence will inherently find a way to influence the world, including creating its own means for embodiment and physical manipulation.

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The last time a biological human was able to grasp all human scientific knowledge was hundreds of years ago.

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once a computer is able to match the subtlety and range of human intelligence, it will necessarily soar past it and then continue its double-exponential ascent. A key question regarding the Singularity is whether the “chicken” (strong AI) or the “egg” (nanotechnology) will come first. In other words, will strong AI lead to full nanotechnology (molecular-manufacturing assemblers that can turn information into physical products), or will full nanotechnology lead to strong AI? The logic of the first premise is that strong AI would imply superhuman AI for the reasons just cited, and superhuman AI would be in a position to solve any remaining design problems required to implement full nanotechnology. The second premise is based on the realization that the hardware requirements for strong AI will be met by nanotechnology-based computation. Likewise the software requirements will be facilitated by nanobots that could create highly detailed scans of human brain functioning and thereby achieve the completion of reverse engineering the human brain. Both premises are logical; it's clear that either technology can assist the other. The reality is that progress in both areas will necessarily use our most advanced tools, so advances in each field will simultaneously facilitate the other. However, I do expect that full MNT will emerge prior to strong AI, but only by a few years (around 2025 for nanotechnology, around 2029 for strong AI).

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Runaway AI. Once strong AI is achieved, it can readily be advanced and its powers multiplied, as that is the fundamental nature of machine abilities. As one strong AI immediately begets many strong AIs, the latter access their own design, understand and improve it, and thereby very rapidly evolve into a yet more capable, more intelligent AI, with the cycle repeating itself indefinitely. Each cycle not only creates a more intelligent AI but takes less time than the cycle before it, as is the nature of technological evolution (or any evolutionary process). The premise is that once strong AI is achieved, it will immediately become a runaway phenomenon of rapidly escalating superintelligence.¹⁶⁰ My own view is only slightly different. The logic of runaway AI is valid, but we still need to consider the timing. Achieving human levels in a machine will not immediately cause a runaway phenomenon.

Highlight (Yellow) | Location 4881

Once we've succeeded in creating a machine that can pass the Turing test (around 2029), the succeeding period will be an era of consolidation in which nonbiological intelligence will make rapid gains. However, the extraordinary expansion contemplated for the Singularity, in which human intelligence is multiplied by billions, won't take place until the mid-2040s (as discussed in chapter 3).

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We are well into the era of "narrow AI," which refers to artificial intelligence that performs a useful and specific function that once required human intelligence to perform, and does so at human levels or better. Often narrow AI systems greatly exceed the speed of humans, as well as provide the ability to manage and consider thousands of variables simultaneously.

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These time frames for AI's technology cycle (a couple of decades of growing enthusiasm, a decade of disillusionment, then a decade and a half of solid advance in adoption) may seem lengthy, compared to the relatively rapid phases of the Internet and telecommunications cycles (measured in years, not decades), but two factors must be considered. First, the Internet and telecommunications cycles were relatively recent, so they are more affected by the acceleration of paradigm shift (as discussed in chapter 1). So recent adoption cycles (boom, bust, and recovery) will be much faster than ones that started forty years ago. Second, the AI revolution is the most profound transformation that human civilization will experience, so it will take longer to mature than less complex technologies. It is characterized by the mastery of the most important and most powerful attribute of human civilization, indeed of the entire sweep of evolution on our planet: intelligence.

Highlight (Yellow) | Location 4923

scientists discovered a subtle property of curved surfaces known as Bernoulli's principle: a gas (such as air) travels more quickly over a curved surface than over a flat surface. Thus, air pressure over a curved surface is lower than over a flat surface. By understanding, focusing, and amplifying the implications of this subtle observation, our engineering created all of aviation. Once we understand the principles of intelligence, we will have a similar opportunity to focus, concentrate, and amplify its powers.

Highlight (Yellow) | Location 4934

An underlying problem with artificial intelligence that I have personally experienced in my forty years in this area is that as soon as an AI technique works, it's no longer considered AI and is spun off as its own field (for example, character recognition, speech recognition, machine vision, robotics, data mining, medical informatics, automated investing).

Highlight (Yellow) | Location 4936

Computer scientist Elaine Rich defines AI as “the study of how to make computers do things at which, at the moment, people are better.” Rodney Brooks, director of the MIT AI Lab, puts it a different way: “Every time we figure out a piece of it, it stops being magical; we say, Oh, that’s just a computation.” I am also reminded of Watson’s remark to Sherlock Holmes, “I thought at first that you had done something clever, but I see that there was nothing in it after all.”¹⁶⁴ That has been our experience as AI scientists. The enchantment of intelligence seems to be reduced to “nothing” when we fully understand its methods. The mystery that is left is the intrigue inspired by the remaining, not as yet understood methods of intelligence.

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in the absence of tools that could peer into the brain with sufficient resolution, AI scientists and engineers developed their own techniques. Just as aviation engineers did not model the ability to fly on the flight of birds, these early AI methods were not based on reverse engineering natural intelligence.

Highlight (Yellow) | Location 4951

Expert Systems. In the 1970s AI was often equated with one specific method: expert systems. This involves the development of specific logical rules to simulate the decision-making processes of human experts.

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Bayesian Nets. Over the last decade a technique called Bayesian logic has created a robust mathematical foundation for combining thousands or even millions of such probabilistic rules in what are called “belief networks” or Bayesian nets. Originally devised by English mathematician Thomas Bayes and published posthumously in 1763, the approach is intended to determine the likelihood of future events based on similar occurrences in the past.¹⁶⁸ Many expert systems based on Bayesian techniques gather data from experience in an ongoing fashion, thereby continually learning and improving their decision making. The most promising type of spam filters are based on this method. I personally use a spam filter called SpamBayes, which trains itself on e-mail that you have identified as either “spam” or “okay.”¹⁶⁹

Highlight (Yellow) | Location 4989

Markov Models. Another method that is good at applying probabilistic networks to complex sequences of information involves Markov models.¹⁷⁰ Andrei Andreyevich Markov (1856–1922), a renowned mathematician, established a theory of “Markov chains,” which was refined by Norbert Wiener (1894–1964) in

1923. The theory provided a method to evaluate the likelihood that a certain sequence of events would occur. It has been popular, for example, in speech recognition, in which the sequential events are phonemes (parts of speech).

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Neural Nets. Another popular self-organizing method that has also been used in speech recognition and a wide variety of other pattern-recognition tasks is neural nets. This technique involves simulating a simplified model of neurons and interneuronal connections.

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Since the neural-net wiring and synaptic weights are initially set randomly, the answers of an untrained neural net will be random. The key to a neural net, therefore, is that it must learn its subject matter. Like the mammalian brains on which it's loosely modeled, a neural net starts out ignorant. The neural net's teacher—which may be a human, a computer program, or perhaps another, more mature neural net that has already learned its lessons—rewards the student neural net when it generates the right output and punishes it when it does not.

Highlight (Yellow) | Location 5029

Genetic Algorithms (GAs). Another self-organizing paradigm inspired by nature is genetic, or evolutionary, algorithms, which emulate evolution, including sexual reproduction and mutations.

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In the novel *usr/bin/god*, Cory Doctorow, a leading science-fiction writer, uses an intriguing variation of a GA to evolve an AI.

Highlight (Yellow) | Location 5073

Recursive Search. Often we need to search through a vast number of combinations of possible solutions to solve a given problem. A classic example is in playing games such as chess.

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the “minimax” algorithm (the standard game-playing method that relies on building trees of move-countermove sequences, and then evaluating the terminal-leaf positions of each branch of the tree) specifically for chess moves.

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It's well known to computer-systems designers that specialized hardware can generally implement a specific algorithm at least one hundred times faster than a general-purpose computer. Specialized ASICs (application-

specific integrated circuits) require significant development efforts and costs, but for critical calculations that are needed on a repetitive basis (for example, decoding MP3 files or rendering graphics primitives for video games), this expenditure can be well worth the investment.

Highlight (Yellow) | Location 5171

Deep Fritz-like programs running on ordinary personal computers will routinely defeat all humans later in this decade. Then we'll really lose interest in chess. Combining Methods. The most powerful approach to building robust AI systems is to combine approaches, which is how the human brain works.

Highlight (Yellow) | Location 5187

Computer science professor and AI entrepreneur Ben Goertzel has written a series of books and articles that describe strategies and architectures for combining the diverse methods underlying intelligence. His Novamente architecture is intended to provide a framework for general-purpose AI.¹⁷⁹

Highlight (Yellow) | Location 5199

In our reporting on the KurzweilAI.net Web site, we feature one or more dramatic systems almost every day.¹⁸⁰

Highlight (Yellow) | Location 5202

a small sample of narrow AI in action. Military and Intelligence. The U.S. military has been an avid user of AI systems. Pattern-recognition software systems guide autonomous weapons such as cruise missiles, which can fly thousands of miles to find a specific building or even a specific window.¹⁸²

Highlight (Yellow) | Location 5209

Expert systems incorporating Bayesian networks and GAs are used to optimize complex supply chains that coordinate millions of provisions, supplies, and weapons based on rapidly changing battlefield requirements.

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Advance warning of the September 11, 2001, terrorist attacks was apparently detected by the National Security Agency's AI-based Echelon system, which analyzes the agency's extensive monitoring of communications traffic.¹⁸⁴ Unfortunately, Echelon's warnings were not reviewed by human agents until it was too late.

Highlight (Yellow) | Location 5216

In the Iraq war that began in 2003 the armed Predator (operated by the CIA) and other flying unmanned aerial vehicles (UAVs) destroyed thousands of enemy tanks and missile sites.

Highlight (Yellow) | Location 5219

Space Exploration. NASA is building self-understanding into the software controlling its unmanned spacecraft.

Highlight (Yellow) | Location 5237

able to detect very subtle phenomena, such as a star blinking for a nanosecond, which may indicate a small asteroid in the outer regions of our solar system passing in front of the light from that star.¹⁸⁷ One such system, called Moving Object and Transient Event Search System (MOTESS), has identified on its own 180 new asteroids and several comets during its first two years of operation. “We have an intelligent observing system,” explained University of Exeter astronomer Alasdair Allan. “It thinks and reacts for itself, deciding whether something it has discovered is interesting enough to need more observations. If more observations are needed, it just goes ahead and gets them.”

Highlight (Yellow) | Location 5246

Medicine. If you obtain an electrocardiogram (ECG) your doctor is likely to receive an automated diagnosis using pattern recognition applied to ECG recordings. My own company (Kurzweil Technologies) is working with United Therapeutics to develop a new generation of automated ECG analysis for long-term unobtrusive monitoring (via sensors embedded in clothing and wireless communication using a cell phone) of the early warning signs of heart disease.¹⁸⁹

Highlight (Yellow) | Location 5258

Men with elevated prostate-specific antigen (PSA) levels typically undergo surgical biopsy, but about 75 percent of these men do not have prostate cancer. A new test, based on pattern recognition of proteins in the blood, would reduce this false positive rate to about 29 percent.¹⁹³ The test is based on an AI program designed by Correlogic Systems in Bethesda, Maryland, and the accuracy is expected to improve further with continued development.

Highlight (Yellow) | Location 5261

Pattern recognition applied to protein patterns has also been used in the detection of ovarian cancer. The best contemporary test for ovarian cancer, called CA-125, employed in combination with ultrasound, misses almost all early-stage tumors. “By the time it is now diagnosed, ovarian cancer is too often deadly,” says Emanuel Petricoin III, codirector of the Clinical Proteomics Program run by the FDA and the National Cancer Institute. Petricoin is the lead developer of a new AI-based test looking for unique patterns of proteins found only in the presence of cancer. In an evaluation involving hundreds of blood samples, the test was, according to Petricoin, “an astonishing 100% accurate in detecting cancer, even at the earliest stages.”¹⁹⁴ About 10 percent of all Pap-smear slides in the United States are analyzed by a self-learning AI program called FocalPoint, developed by TriPath Imaging.

Highlight (Yellow) | Location 5275

Science and Math.

Highlight (Yellow) | Location 5284

A long-standing conjecture in algebra was finally proved by an AI system at Argonne National Laboratory. Human mathematicians called the proof “creative.”

Highlight (Yellow) | Location 5285

Business, Finance, and Manufacturing. Companies in every industry are using AI systems to control and optimize logistics, detect fraud and money laundering, and perform intelligent data mining on the horde of information they gather each day.

Highlight (Yellow) | Location 5290

Future Route, an English company, for example, offers iHex, based on AI routines developed at Oxford University, to detect fraud in credit-card transactions and loan applications.¹⁹⁸ The system continuously generates and updates its own rules based on its experience. First Union Home Equity Bank in Charlotte, North Carolina, uses Loan Arranger, a similar AI-based system, to decide whether to approve mortgage applications.¹⁹⁹

Highlight (Yellow) | Location 5294

NASDAQ similarly uses a learning program called the Securities Observation, News Analysis, and Regulation (SONAR) system to monitor all trades for fraud as well as the possibility of insider trading.²⁰⁰ As of the end of 2003 more than 180 incidents had been detected by SONAR and referred to the U.S. Securities and Exchange Commission and Department of Justice. These included several cases that later received significant news coverage.

Highlight (Yellow) | Location 5297

Ascent Technology, founded by Patrick Winston, who directed MIT’s AI Lab from 1972 through 1997, has designed a GA-based system called Smart-Airport Operations Center (SAOC) that can optimize the complex logistics of an airport, such as balancing work assignments of hundreds of employees, making gate and equipment assignments, and managing a myriad of other details.²⁰¹

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Armando Fox, the head of Stanford University’s Software Infrastructures Group, who is working on what is now called “autonomic computing.” Fox adds, “The system has to be able to set itself up, it has to optimize itself. It has to repair itself, and if something goes wrong, it has to know how to respond to external threats.”

Highlight (Yellow) | Location 5325

in a single pass build a reliable “voxel” (three-dimensional pixel) map of the environment. The robot can then use the map and its own reasoning ability to determine an optimal and obstacle-free path to carry out its assigned mission.

Highlight (Yellow) | Location 5331

a key element for natural interactions. Head- and eye-tracking systems have been developed at Carnegie Mellon University and MIT

Highlight (Yellow) | Location 5340

Speech and Language. Dealing naturally with language is the most challenging task of all for artificial intelligence.

Highlight (Yellow) | Location 5349

Microsoft has developed a natural-language search engine called Ask MSR (Ask Micro-Soft Research), which actually answers natural-language questions such as “When was Mickey Mantle born?”²⁰⁶

Highlight (Yellow) | Location 5364

Not every caller is satisfied with the ability of these virtual agents to get the job done, but most systems provide a means to get a human on the line. Companies using these systems report that they reduce the need for human service agents up to 80 percent.

Highlight (Yellow) | Location 5369

In 2005 the Acura RL and Honda Odyssey will be offering a system from IBM that allows users to converse with their cars.²⁰⁸ Driving directions will include street names (for example, “turn left on Main Street, then right on Second Avenue”). Users can ask such questions as “Where is the nearest Italian restaurant?”

Highlight (Yellow) | Location 5372

The Acura RL will also track road conditions and highlight traffic congestion on its screen in real time.

Highlight (Yellow) | Location 5377

Franz Josef Och, a computer scientist at the University of Southern California, has developed a technique that can generate a new language-translation system between any pair of languages in a matter of hours or days.²⁰⁹

Highlight (Yellow) | Location 5403

when someone learns “by rote,” we say that they don’t really understand. However, if you have several different representations then, when one approach fails you can try another.

Highlight (Yellow) | Location 5408

Advancing computer performance is like water slowly flooding the landscape. A half century ago it began to drown the lowlands, driving out human calculators and record clerks, but leaving most of us dry. Now the flood

has reached the foothills, and our outposts there are contemplating retreat. We feel safe on our peaks, but, at the present rate, those too will be submerged within another half century. I propose that we build Arks as that day nears, and adopt a seafaring life! For now, though, we must rely on our representatives in the lowlands to tell us what water is really like.

Highlight (Yellow) | Location 5415

After all, the machines were doing something beyond any animal, that needed human intelligence, concentration and years of training. But it is hard to recapture that magic now. One reason is that computers' demonstrated stupidity in other areas biases our judgment.

Highlight (Yellow) | Location 5446

The three tasks that have to do with human-level understanding of natural language—reviewing a movie, holding a press conference, and translating speech—are the most difficult. Once we can take down these signs, we'll have Turing-level machines, and the era of strong AI will have started.

Highlight (Yellow) | Location 5456

just taking the semiconductor industry's published ITRS road map, which runs to 2018, we can project human-level hardware at reasonable cost by that year.²¹⁶ I've stated the case in chapter 4 of why we can have confidence that we will have detailed models and simulations of all regions of the human brain by the late 2020s.

Highlight (Yellow) | Location 5475

the brain differs in a number of important ways from conventional, contemporary computers. If you open up your Palm Pilot and cut a wire, there's a good chance you will break the machine. Yet we routinely lose many neurons and interneuronal connections with no ill effect, because the brain is self-organizing and relies on distributed patterns in which many specific details are not important. When we get to the mid- to late 2020s, we will have access to a generation of extremely detailed brain-region models.

Highlight (Yellow) | Location 5481

Part of the brain's strategy is to learn information, rather than having knowledge hard-coded from the start. ("Instinct" is the term we use to refer to such innate knowledge.) Learning will be an important aspect of AI, as well. In my experience in developing pattern-recognition systems in character recognition, speech recognition, and financial analysis, providing for the AI's education is the most challenging and important part of the engineering. With the accumulated knowledge of human civilization increasingly accessible online, future AIs will have the opportunity to conduct their education by accessing this vast body of information.

Highlight (Yellow) | Location 5486

The education of AIs will be much faster than that of unenhanced humans. The twenty-year time span required to provide a basic education to biological humans could be compressed into a matter of weeks or less. Also,

because nonbiological intelligence can share its patterns of learning and knowledge, only one AI has to master each particular skill.

Highlight (Yellow) | Location 5495

Variations of the Turing test have been proposed. The annual Loebner Prize contest awards a bronze prize to the chatterbot (conversational bot) best able to convince human judges that it's human.²¹⁷ The criteria for winning the silver prize is based on Turing's original test, and it obviously has yet to be awarded. The gold prize is based on visual and auditory communication.

Highlight (Yellow) | Location 5504

Turing was carefully imprecise in setting the rules for his test, and significant literature has been devoted to the subtleties of establishing the exact procedures for determining how to assess when the Turing test has been passed.²¹⁸

Highlight (Yellow) | Location 5509

Simply defining "machine" and "human," for example, was not a straightforward matter. Is the human judge allowed to have any nonbiological thinking processes in his or her brain? Conversely, can the machine have any biological aspects?

Highlight (Yellow) | Location 5513

By the time there is a broad consensus that the Turing test has been passed, the actual threshold will have long since been achieved.

Highlight (Yellow) | Location 5514

Edward Feigenbaum proposes a variation of the Turing test, which assesses not a machine's ability to pass for human in casual, everyday dialogue but its ability to pass for a scientific expert in a specific field.²²⁰ The Feigenbaum test (FT) may be more significant than the Turing test because FT-capable machines, being technically proficient, will be capable of improving their own designs.

Highlight (Yellow) | Location 5527

history of AI reveals that machines started with the skills of professionals and only gradually moved toward the language skills of a child.

Highlight (Yellow) | Location 5533

Passing the FT in all disciplines is likely to take longer, however. This is why I see the 2030s as a period of consolidation, as machine intelligence rapidly expands its skills and incorporates the vast knowledge bases of our biological human and machine civilization. By the 2040s we will have the opportunity to apply the

accumulated knowledge and skills of our civilization to computational platforms that are billions of times more capable than unassisted biological human intelligence.

Highlight (Yellow) | Location 5537

advent of strong AI is the most important transformation this century will see.

Highlight (Yellow) | Location 5538

It will mean that a creation of biology has finally mastered its own intelligence and discovered means to overcome its limitations.

Highlight (Yellow) | Location 5542

Intelligence is the ability to solve problems with limited resources, including limitations of time. The Singularity will be characterized by the rapid cycle of human intelligence—increasingly nonbiological—capable of comprehending and leveraging its own powers.

Highlight (Yellow) | Location 5544

FRIEND OF FUTURIST BACTERIUM, 2 BILLION B.C.: So tell me again about these ideas you have about the future. FUTURIST BACTERIUM, 2 BILLION B.C.: Well, I see bacteria getting together into societies, with the whole band of cells basically acting like one big complicated organism with greatly enhanced capabilities. FRIEND OF FUTURIST BACTERIUM: What gives you that idea? FUTURIST BACTERIUM: Well already, some of our fellow Daptobacters have gone inside other larger bacteria to form a little duo.²²¹ It's inevitable that our fellow cells will band together so that each cell can specialize its function. As it is now, we each have to do everything by ourselves: find food, digest it, excrete by-products. FRIEND OF FUTURIST BACTERIUM: And then what? FUTURIST BACTERIUM: All these cells will develop ways of communicating with one another that go beyond just the swapping of chemical gradients that you and I can do. FRIEND OF FUTURIST BACTERIUM: Okay, now tell me again the part about that future superassembly often trillion cells. FUTURIST BACTERIUM: Yes, well, according to my models, in about two billion years a big society often trillion cells will make up a single organism and include tens of billions of special cells that can communicate with one another in very complicated patterns. FRIEND OF FUTURIST BACTERIUM: What sort of patterns? FUTURIST BACTERIUM: Well, "music," for one thing. These huge bands of cells will create musical patterns and communicate them to all the other bands of cells. FRIEND OF FUTURIST BACTERIUM: Music? FUTURIST BACTERIUM: Yes, patterns of sound. FRIEND OF FUTURIST BACTERIUM: Sound? FUTURIST BACTERIUM: Okay, look at it this way. These supercell societies will be complicated enough to understand their own organization. They will be able to improve their own design, getting better and better, faster and faster. They will reshape the rest of the world in their image. FRIEND OF FUTURIST BACTERIUM: Now, wait a second. Sounds like we'll lose our basic bacteriumity. FUTURIST BACTERIUM: Oh, but there will be no loss. FRIEND OF FUTURIST BACTERIUM: I know you keep saying that, but ... FUTURIST BACTERIUM: It will be a great step forward. It's our destiny as bacteria. And, anyway, there will still be little bacteria like us floating around. FRIEND OF FUTURIST BACTERIUM: Okay, but what about the downside? I mean, how much harm can our fellow Daptobacter and Bdellovibrio bacteria do? But these future cell

associations with their vast reach may destroy everything. FUTURIST BACTERIUM: It's not certain, but I think we'll make it through. FRIEND OF FUTURIST BACTERIUM: You always were an optimist. FUTURIST BACTERIUM: Look, we won't have to worry about the downside for a couple billion years. FRIEND OF FUTURIST BACTERIUM: Okay, then, let's get lunch. MEANWHILE, TWO BILLION YEARS LATER ... NED LUDD: These future intelligences will be worse than the textile machines I fought back in 1812. Back then we had to worry about only one man with a machine doing the work of twelve. But you're talking about a marble-size machine outperforming all of humanity. RAY: It will only outperform the biological part of humanity. In any event, that marble is still human, even if not biological. NED: These super intelligences won't eat food. They won't breathe air. They won't reproduce through sex So just how are they human? RAY: We're going to merge with our technology. We're already starting to do that in 2004, even if most of the machines are not yet inside our bodies and brains. Our machines nonetheless extend the reach of our intelligence. Extending our reach has always been the nature of being human. NED: Look, saying that these superintelligent nonbiological entities are human is like saying that we're basically bacteria. After all, we're evolved from them also. RAY: It's true that a contemporary human is a collection of cells, and that we are a product of evolution, indeed its cutting edge. But extending our intelligence by reverse engineering it, modeling it, simulating it, reinstantiating it on more capable substrates, and modifying and extending it is the next step in its evolution. It was the fate of bacteria to evolve into a technology-creating species. And it's our destiny now to evolve into the vast intelligence of the Singularity.

Highlight (Yellow) | Location 5607

"Playing God" is actually the highest expression of human nature. The urges to improve ourselves, to master our environment, and to set our children on the best path possible have been the fundamental driving forces of all of human history. Without these urges to "play God," the world as we know it wouldn't exist today. A few million humans would live in savannahs and forests, eking out a hunter-gatherer existence, without writing or history or mathematics or an appreciation of the intricacies of their own universe and their own inner workings. —RAMEZ NAAM

Highlight (Yellow) | Location 5618

the intertwined revolutions of G, N, and R will transform our frail version 1.0 human bodies into their far more durable and capable version 2.0 counterparts. Billions of nanobots will travel through the bloodstream in our bodies and brains. In our bodies, they will destroy pathogens, correct DNA errors, eliminate toxins, and perform many other tasks to enhance our physical well-being. As a result, we will be able to live indefinitely without aging. In our brains, the massively distributed nanobots will interact with our biological neurons. This will provide full-immersion virtual reality incorporating all of the senses, as well as neurological correlates of our emotions, from within the nervous system. More important, this intimate connection between our biological thinking and the nonbiological intelligence we are creating will profoundly expand human intelligence. Warfare will move toward nanobot-based weapons, as well as cyber-weapons. Learning will first move online, but once our brains are online we will be able to download new knowledge and skills. The role of work will be to create knowledge of all kinds, from music and art to math and science. The role of play will be, well, to create knowledge, so there won't be a clear distinction between work and play. Intelligence on and around the Earth will continue to expand exponentially until we reach the limits of matter and energy to support intelligent computation. As we approach this limit in our corner of the galaxy, the intelligence of our civilization will

expand outward into the rest of the universe, quickly reaching the fastest speed possible. We understand that speed to be the speed of light, but there are suggestions that we may be able to circumvent this apparent limit (possibly by taking shortcuts through wormholes, for example).

Highlight (Yellow) | Location 5645

A New Way of Eating. Sex has largely been separated from its biological function. For the most part, we engage in sexual activity for intimate communication and sensual pleasure, not reproduction.

Highlight (Yellow) | Location 5649

So why don't we provide the same extrication of purpose from biology for another activity that also provides both social intimacy and sensual pleasure—namely, eating?

Highlight (Yellow) | Location 5667

We now live in an era of great material abundance, at least in technologically advanced nations. Most work requires mental effort rather than physical exertion. A century ago 30 percent of the U.S. workforce was employed on farms, with another 30 percent in factories. Both of these figures are now under 3 percent.² Many of today's job categories, ranging from flight controller to Web designer, simply didn't exist a century ago.

Highlight (Yellow) | Location 5673

Our species has already augmented our natural lifespan through our technology: drugs, supplements, replacement parts for virtually all bodily systems, and many other interventions. We have devices to replace our hips, knees, shoulders, elbows, wrists, jaws, teeth, skin, arteries, veins, heart valves, arms, legs, feet, fingers, and toes, and systems to replace more complex organs (for example, our hearts) are beginning to be introduced. As we learn the operating principles of the human body and brain, we will soon be in a position to design vastly superior systems that will last longer and perform better, without susceptibility to breakdown, disease, and aging.

Highlight (Yellow) | Location 5686

next phase of improvement in this area will be largely biochemical, in the form of drugs and supplements that will prevent excess caloric absorption and otherwise reprogram metabolic pathways for optimal health. Research by Dr. Ron Kahn at the Joslin Diabetes Center has already identified the "fat insulin receptor" (FIR) gene, which controls accumulation of fat by the fat cells. By blocking the expression of this single gene in the fat cells of mice, Dr. Kahn's pioneering research has demonstrated that the animals were able to eat without restriction yet remain lean and healthy. Although they ate far more than the control mice, the "FIR knockout" mice actually lived 18 percent longer and had substantially lower rates of heart disease and diabetes.

Highlight (Yellow) | Location 5698

nanoengineered blood-borne devices that deliver hormones such as insulin have been demonstrated in animals.⁶ Similar systems could precisely deliver dopamine to the brain for Parkinson's patients, provide blood-clotting

factors for patients with hemophilia, and deliver cancer drugs directly to tumor sites. One new design provides up to twenty substance-containing reservoirs that can release their cargo at programmed times and locations in the body.⁷

Highlight (Yellow) | Location 5702

Kensall Wise, a professor of electrical engineering at the University of Michigan, has developed a tiny neural probe that can provide precise monitoring of the electrical activity of patients with neural diseases.⁸ Future designs are also expected to deliver drugs to precise locations in the brain.

Highlight (Yellow) | Location 5722

As the nutrient nanobots make their way into our bodies, the elimination nanobots go the other way. Such an innovation would also enable us to out-grow the need for the organs that filter the blood for impurities, such as the kidneys.

Highlight (Yellow) and Note | Location 5728

Of course, most of us won't do away with our old-fashioned digestive process when these technologies are first introduced. After all, people didn't throw away their typewriters when the first generation of word processors was introduced. However, these new technologies will in due course dominate. Few people today still use a typewriter, a horse and buggy, a wood-burning stove, or other displaced technologies (other than as deliberate experiences in antiquity).

Amish & Chabader's stand out as the oddity for shunning 'reasonable' progress 100yrs later...

Highlight (Yellow) | Location 5732

Once we've worked out the inevitable complications that will arise with a radically reengineered gastrointestinal system, we'll begin to rely on it more and more. A nanobot-based digestive system can be introduced gradually, first augmenting our digestive tract, replacing it only after many iterations.

Highlight (Yellow) and Note | Location 5737

Freitas has redesigned them for optimal performance. Because his respirocites (robotic red blood cells) would enable one to go hours without oxygen,¹²

a next gen twist on SCUBA

Highlight (Yellow) | Location 5739

Presumably the use of respirocites and similar systems will be prohibited in events like the Olympics, but then we will face the prospect of teenagers (whose bloodstreams will likely contain respirocite-enriched blood) routinely outperforming Olympic athletes.

Highlight (Yellow) | Location 5751

approach will be to get rid of the heart altogether. Among Freitas's designs are nanorobotic blood cells that provide their own mobility.

Highlight (Yellow) | Location 5758

With the respirocytes providing greatly improved oxygenation, we will be able to eliminate the lungs by using nanobots to provide oxygen and remove carbon dioxide. As with other systems, we will go through intermediate stages where these technologies simply augment our natural processes, so we can have the best of both worlds. Eventually, though, there will be no reason to continue with the complications of actual breathing and the burdensome requirement of breathable air everywhere we go.

Highlight (Yellow) | Location 5769

Since we will be eliminating most of our biological organs, many of these substances may no longer be needed and will be replaced by other resources required by the nanorobotic systems. So What's Left? Let's consider where we are, circa early 2030s. We've eliminated the heart, lungs, red and white blood cells, platelets, pancreas, thyroid and all the hormone-producing organs, kidneys, bladder, liver, lower esophagus, stomach, small intestines, large intestines, and bowel. What we have left at this point is the skeleton, skin, sex organs, sensory organs, mouth and upper esophagus, and brain.

Highlight (Yellow) | Location 5776

Interlinking nanobots will one day provide the ability to augment and ultimately replace the skeleton through a gradual and noninvasive process. The human skeleton version 2.0 will be very strong, stable, and self-repairing.

Highlight (Yellow) | Location 5777

We will not notice the absence of many of our organs, such as the liver and pancreas, since we do not directly experience their operation. But the skin, which includes our primary and secondary sex organs, may prove to be an organ we will actually want to keep,

Highlight (Yellow) | Location 5779

we may at least want to maintain its vital functions of communication and pleasure. However, we will ultimately be able to improve on the skin with new nanoengineered supple materials that will provide greater protection from physical and thermal environmental effects while enhancing our capacity for intimate communication. The same observation holds for the mouth and upper esophagus, which constitute the remaining aspects of the digestive system that we use to experience the act of eating.

Highlight (Yellow) | Location 5788

An implant for people with cerebral palsy and multiple sclerosis communicates with the ventral lateral thalamus and has been effective in controlling tremors.²¹ "Rather than treat the brain like soup, adding chemicals that

enhance or suppress certain neurotransmitters,” says Rick Trosch, an American physician helping to pioneer these therapies, “we’re now treating it like circuitry.”

Highlight (Yellow) | Location 5793

Researchers at Germany’s Max Planck Institute have developed noninvasive devices that can communicate with neurons in both directions.²² They demonstrated their “neuron transistor” by controlling the movements of a living leech from a personal computer. Similar technology has been used to reconnect leech neurons and coax them to perform simple logical and arithmetic problems.

Highlight (Yellow) | Location 5805

We will also be able to place sensors in the brain of a paralyzed person that will be programmed to recognize the brain patterns associated with intended movements and then stimulate the appropriate sequence of muscle actions.

Highlight (Yellow) | Location 5809

We Are Becoming Cyborgs. The human body version 2.0 scenario represents the continuation of a long-standing trend in which we grow more intimate with our technology. Computers started out as large, remote machines in air-conditioned rooms tended by white-coated technicians. They moved onto our desks, then under our arms, and now into our pockets. Soon, we’ll routinely put them inside our bodies and brains. By the 2030s we will become more nonbiological than biological. As I discussed in chapter 3, by the 2040s nonbiological intelligence will be billions of times more capable than our biological intelligence.

Highlight (Yellow) | Location 5815

Stephen Hawking recently commented in the German magazine *Focus* that computer intelligence will surpass that of humans within a few decades. He advocated that we “urgently need to develop direct connections to the brain, so that computers can add to human intelligence, rather than be in opposition.”²⁵ Hawking can take comfort that the development program he is recommending is well under way.

Highlight (Yellow) | Location 5820

Biological evolution is only capable of what is called “local optimization,” meaning that it can improve a design but only within the constraints of design “decisions” that biology arrived at long ago. For example, biological evolution is restricted to building everything from a very limited class of materials—namely, proteins, which are folded from one-dimensional strings of amino acids. It is restricted to thinking processes (pattern recognition, logical analysis, skill formation, and other cognitive skills) that use extremely slow chemical switching. And biological evolution itself works very slowly, only incrementally improving designs that continue to apply these basic concepts. It is incapable of suddenly changing, for example, to structural materials made of diamondoid or to nanotube-based logical switching.

Highlight (Yellow) | Location 5828

Human Body Version 3.0. I envision human body 3.0—in the 2030s and 2040s—as a more fundamental redesign. Rather than reformulating each subsystem, we (both the biological and nonbiological portions of our thinking, working together) will have the opportunity to revamp our bodies based on our experience with version 2.0. As with the transition from 1.0 to 2.0, the transition to 3.0 will be gradual and will involve many competing ideas. One attribute I envision for version 3.0 is the ability to change our bodies. We’ll be able to do that very easily in virtual-reality environments (see the next section), but we will also acquire the means to do this in real reality. We will incorporate MNT-based fabrication into ourselves, so we’ll be able to rapidly alter our physical manifestation at will. Even with our mostly nonbiological brains we’re likely to keep the aesthetics and emotional import of human bodies, given the influence this aesthetic has on the human brain. (Even when extended, the nonbiological portion of our intelligence will still have been derived from biological human intelligence.) That is, human body version 3.0 is likely still to look human by today’s standards, but given the greatly expanded plasticity that our bodies will have, ideas of what constitutes beauty will be expanded upon over time. Already, people augment their bodies with body piercing, tattoos, and plastic surgery,

Highlight (Yellow) | Location 5839

J. Storrs Hall has described nanobot designs he calls “foglets” that are able to link together to form a great variety of structures and that can quickly change their structural organization. They’re called “foglets” because if there’s a sufficient density of them in an area, they can control sound and light to form variable sounds and images. They are essentially creating virtual-reality environments externally (that is, in the physical world) rather than internally (in the nervous system). Using them a person can modify his body or his environment, though some of these changes will actually be illusions, since the foglets can control sound and images.²⁶

Highlight (Yellow) | Location 5862

BILL: Better only in the sense of replacing diseased or disabled organs and systems. But you’re replacing essentially all of our humanness to enhance human ability, and that’s inherently inhuman. RAY: Then perhaps our basic disagreement is over the nature of being human. To me, the essence of being human is not our limitations—although we do have many—it’s our ability to reach beyond our limitations. We didn’t stay on the ground. We didn’t even stay on the planet. And we are already not settling for the limitations of our biology.

Highlight (Yellow) | Location 5877

One windy day two monks were arguing about a flapping banner. The first said, “I say the banner is moving, not the wind.” The second said, “I say the wind is moving, not the banner.” A third monk passed by and said, “The wind is not moving. The banner is not moving. Your minds are moving.” —ZEN PARABLE

Highlight (Yellow) | Location 5883

The 2010 Scenario. Computers arriving at the beginning of the next decade will become essentially invisible: woven into our clothing, embedded in our furniture and environment. They will tap into the worldwide mesh (what the World Wide Web will become once all of its linked devices become communicating Web servers,

thereby forming vast supercomputers and memory banks) of high-speed communications and computational resources.

Highlight (Yellow) | Location 5899

your retinal display might remind us, “That’s Dr. John Smith, director of the ABC Institute—you last saw him six months ago at the XYZ conference” or, “That’s the Time-Life Building—your meeting is on the tenth floor.” We’ll have real-time translation of foreign languages, essentially subtitles on the world, and access to many forms of online information integrated into our daily activities.

Highlight (Yellow) | Location 5903

These virtual assistants won’t always wait for questions and directives but will step forward if they see us struggling to find a piece of information.

Highlight (Yellow) and Note | Location 5909

For example, scientists at the Max Planck Institute have developed “neuron transistors” that can detect the firing of a nearby neuron, or alternatively can cause a nearby neuron to fire or suppress it from firing.³¹

repeating himself...

Highlight (Yellow) | Location 5916

discussed in chapter 4, inputs from the body—comprising a few hundred megabits per second—representing information about touch, temperature, acid levels, the movement of food, and other physical events, stream through the Lamina 1 neurons, then through the posterior ventromedial nucleus, ending up in the two insula regions of cortex. If these are coded correctly—and we will know how to do that from the brain reverse-engineering effort—your brain will experience the synthetic signals just as it would real ones.

Highlight (Yellow) | Location 5924

We will be able to visit these virtual places and have any kind of interaction with other real, as well as simulated, people (of course, ultimately there won’t be a clear distinction between the two), ranging from business negotiations to sensual encounters. “Virtual-reality environment designer” will be a new job description and a new art form.

Highlight (Yellow) | Location 5958

Experience Beamers. “Experience beamers” will send the entire flow of their sensory experiences as well as the neurological correlates of their emotional reactions out onto the Web, just as people today beam their bedroom images from their Web cams. A popular pastime will be to plug into someone else’s sensory-emotional beam and experience what it’s like to be that person, à la the premise of the movie Being John Malkovich.

Highlight (Yellow) | Location 5962

Expand Your Mind. The most important application of circa-2030 nanobots will be literally to expand our minds through the merger of biological and nonbiological intelligence.

Highlight (Yellow) | Location 5964

The first stage will be to augment our hundred trillion very slow interneuronal connections with high-speed virtual connections via nanorobot communication.³⁶ This will provide us with the opportunity to greatly boost our pattern-recognition abilities, memories, and overall thinking capacity, as well as to directly interface with powerful forms of nonbiological intelligence. The technology will also provide wireless communication from one brain to another.

Highlight (Yellow) | Location 5988

Nanobots will be introduced without surgery, through the bloodstream, and if necessary can all be directed to leave, so the process is easily reversible. They are programmable, in that they can provide virtual reality one minute and a variety of brain extensions the next.

Highlight (Yellow) | Location 5994

you wont feel them, any more than you feel the neurons in your head or the bacteria in your GI tract.

Highlight (Yellow) | Location 5996

SIGMUND FREUD: Hmmm, that's what they used to say about the telephone when I was young. People would say, "Who needs to talk to someone hundreds of miles away when you can just get together?" RAY: Exactly, the telephone is auditory virtual reality. So full-immersion VR is, basically, a full-body telephone. You can get together with anyone anytime but do more than just talk.

Highlight (Yellow) | Location 6000

GEORGE 2048: It's certainly been a boon for sex workers; they never have to leave their homes. It became so impossible to draw any meaningful lines that the authorities had no choice but to legalize virtual prostitution in 2033.

Highlight (Yellow) | Location 6008

RAY: That's another benefit of virtual reality circa 2029; you have your choice of millions of artificial people. MOLLY 2104: I understand that you're back in 2004, but we kind of got rid of that terminology back when the Nonbiological Persons Act was passed in 2052. I mean, we're a lot more real than ... umm, let me rephrase that.

Highlight (Yellow) | Location 6112

As we move toward a nonbiological existence, we will gain the means of “backing ourselves up” (storing the key patterns underlying our knowledge, skills, and personality), thereby eliminating most causes of death as we know it.

Highlight (Yellow) | Location 6115

Life Expectancy (Years)40 Cro-Magnon Era 18 Ancient Egypt 25 1400 Europe 30 1800 Europe and United States 37 1900 United States 48 2002 United States 78

Highlight (Yellow) | Location 6119

A mind that stays at the same capacity cannot live forever; after a few thousand years it would look more like a repeating tape loop than a person. To live indefinitely long, the mind itself must grow, ... and when it becomes great enough, and looks back ... what fellow feeling can it have with the soul that it was originally? The later being would be everything the original was, but vastly more. —VERNOR VINCE

Highlight (Yellow) | Location 6124

I reported on brain uploading in chapter 4. The straightforward brain-porting scenario involves scanning a human brain (most likely from within), capturing all of the salient details, and reinstantiating the brain’s state in a different—most likely much more powerful—computational substrate. This will be a feasible procedure and will happen most likely around the late 2030s. But this is not the primary way that I envision the transition to nonbiological experience taking place. It will happen, rather, in the same way that all other paradigm shifts happen: gradually (but at an accelerating pace).

Highlight (Yellow) | Location 6146

if we are diligent in maintaining our mind file, making frequent backups, and porting to current formats and mediums, a form of immortality can be attained, at least for software-based humans. Later in this century it will seem remarkable to people that humans in an earlier era lived their lives without a backup of their most precious information: that contained in their brains and bodies.

Highlight (Yellow) | Location 6153

But is that person based on my mind file, who migrates across many computational substrates and who outlives any particular thinking medium, really me? This consideration takes us back to the same questions of consciousness and identity that have been debated since Plato’s dialogues (which we examine in the next chapter). During the course of the twenty-first century these will not remain topics for polite philosophical debates but will have to be confronted as vital, practical, political, and legal issues.

Highlight (Yellow) | Location 6170

Do not go gentle into that good night, ... Rage, rage against the dying of the light. —DYLAN THOMAS

Highlight (Yellow) | Location 6216

we are writing our precious heritage in what medical informatics expert Bryan Bergeron calls “disappearing ink,”

Highlight (Yellow) | Location 6224

Copying our minds to other mediums raises a number of philosophical issues, which I will discuss in the next chapter—for example, “Is that really me or rather someone else who just happens to have mastered all my thoughts and knowledge?”

Highlight (Yellow) | Location 6228

For eons the longevity of our mental software has been inexorably linked to the survival of our biological hardware.

Highlight (Yellow) | Location 6232

what can we conclude about the ultimate longevity of software? The answer is simply this: Information lasts only so long as someone cares about it.

Highlight (Yellow) | Location 6233

there is no set of hardware and software standards existing today, nor any likely to come along, that will provide any reasonable level of confidence that the stored information will still be accessible (without unreasonable levels of effort) decades from now.⁴⁵

Highlight (Yellow) | Location 6237

If an archive remains ignored, it will ultimately become as inaccessible as my old eight-inch PDP-8 floppy disks.

Highlight (Yellow) | Location 6257

The great battles of World Wars I and II and the Korean War, in which tens of thousands of lives were lost over the course of a few days, were visually recorded only by occasional grainy newsreels. Today, we have a front-row seat for almost every engagement. Each war has its complexities, but the overall movement toward precision intelligent warfare is clear by examining the number of casualties. This trend is similar to what we are beginning to see in medicine, where smart weapons against disease are able to perform specific missions with far fewer side effects. The trend is similar for collateral casualties, although it may not seem that way from contemporary media coverage (recall that about fifty million civilians died in World War II).

Highlight (Yellow) | Location 6283

New lightweight yet ultrastrong nanomaterials (such as plastics combined with nanotubes, which are fifty times stronger than steel),

Highlight (Yellow) | Location 6295

The FCS is not a one-shot program; it represents a pervasive focus of military systems toward remotely guided, autonomous, miniaturized, and robotic systems, combined with robust, self-organizing, distributed, and secure communications.

Highlight (Yellow) | Location 6307

Swarm intelligence describes the way that complex behaviors can arise from large numbers of individual agents, each following relatively simple rules.⁵² Swarms of insects are often able to devise intelligent solutions to complex problems, such as designing the architecture of a colony, despite the fact that no single member of the swarm possesses the requisite skills.

Highlight (Yellow) | Location 6320

The generals are moving even farther away. Tommy Franks conducted the war in Afghanistan from his bunker in Qatar.

Highlight (Yellow) | Location 6321

Smart Dust. DARPA is developing devices even tinier than birds and bumblebees called “smart dust”—complex sensor systems not much bigger than a pinhead. Once fully developed, swarms of millions of these devices could be dropped into enemy territory to provide highly detailed surveillance and ultimately support offensive

Highlight (Yellow) | Location 6335

University of Illinois has developed self-healing plastics that incorporate microspheres of liquid monomers and a catalyst into a plastic matrix; when a crack appears, the microspheres break, automatically sealing the crack.⁵⁴

Highlight (Yellow) | Location 6347

By the late 2030s and 2040s, as we approach human body version 3.0 and the predominance of nonbiological intelligence, the issue of cyberwarfare will move to center stage. When everything is information, the ability to control your own information and disrupt your enemy’s communication, command, and control will be a primary determinant of military success.

Highlight (Yellow) | Location 6350

Science is organized knowledge. Wisdom is organized life. —IMMANUEL KANT (1724–1804)

Highlight (Yellow) | Location 6352

Most education in the world today, including in the wealthier communities, is not much changed from the model offered by the monastic schools of fourteenth-century Europe.

Highlight (Yellow) | Location 6356

As with all of our other institutions we will ultimately move toward a decentralized educational system in which every person will have ready access to the highest-quality knowledge and instruction. We are now in the early stages of this transformation,

Highlight (Yellow) | Location 6361

MIT offers nine hundred of its courses—half of all its course offerings—for free on the Web.⁵⁶

Highlight (Yellow) | Location 6406

Technology typically starts out with unaffordable products that don't work very well, followed by expensive versions that work a bit better, and then by inexpensive products that work reasonably well. Finally the technology becomes highly effective, ubiquitous, and almost free.

Highlight (Yellow) | Location 6409

Today the delay between early and late adoption is about a decade, but in keeping with the doubling of the paradigm-shift rate every decade, this delay will be only about five years in the middle of the second decade and only a couple of years in the mid-2020s.

Highlight (Yellow) | Location 6414

With the advent of MNT-based manufacturing, the cost of making any physical product will be reduced to pennies per pound, plus the cost of the information guiding the process,

Highlight (Yellow) | Location 6424

Clearly, existing or new business models that allow for the creation of valuable intellectual property (IP) need to be protected, otherwise the supply of IP will itself be threatened.

Highlight (Yellow) | Location 6429

The mobile-phone sector is a prime example of an industry that has not invited rampant piracy. The cost of cell-phone calls has fallen rapidly with improving technology.

Highlight (Yellow) | Location 6437

Each industry will need to continually reinvent its business models, which will require as much creativity as the creation of the IP itself.

Highlight (Yellow) | Location 6438

The first industrial revolution extended the reach of our bodies, and the second is extending the reach of our minds.

Highlight (Yellow) | Location 6446

Decentralization. The next several decades will see a major trend toward decentralization.

Highlight (Yellow) | Location 6449

The ability to do nearly anything with anyone from anywhere in any virtual-reality environment will make obsolete the centralized technologies of office buildings and cities.

Highlight (Yellow) | Location 6465

Einstein put aside his work for the Swiss patent office and engaged in playful mind experiments, resulting in the creation of his enduring theories of special and general relativity. If war is the father of invention, then play is its mother.

Highlight (Yellow) | Location 6473

Although the environments will not be completely convincing at first, by the late 2020s they will be indistinguishable from real reality and will involve all of the senses, as well as neurological correlations of our emotions. As we enter the 2030s there won't be clear distinctions between human and machine, between real and virtual reality, or between work and play.

Highlight (Yellow) | Location 6477

The universe is not only queerer than we suppose, but queerer than we can suppose. —J. B. S. HALDANE

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Kardashev describes a "type II" civilization as one that has harnessed the power of its star for communication using electromagnetic radiation (about 4×10^{26} watts, based on our sun).⁶⁷

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Kardashev's type III: a civilization that has harnessed the energy of its galaxy (about 4×10^{37} watts, based on our galaxy).

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Drake formula, which states: The number of radio-transmitting civilizations = $N \times f_p \times n_e \times f_l \times f_i \times f_c \times f_L$ where: N = the number of stars in the Milky Way galaxy. Current estimates are around 100 billion (1011). f_p =

the fraction of stars that have orbiting planets. Current estimates range from about 20 percent to 50 percent. n_e : For each star with orbiting planets, what is the average number of planets capable of sustaining life? This factor is highly controversial. Some estimates are one or higher (that is, every star with planets has, on average, at least one planet that can sustain life) to much lower factors, such as one in one thousand or even less. f_l : For the planets capable of sustaining life, on what fraction of these does life actually evolve? Estimates are all over the map, from approximately 100 percent to about 0 percent. f_i : For each planet on which life evolves, what is the fraction on which intelligent life evolves? f_l and f_i are the most controversial factors in the Drake equation. Here again, estimates range from nearly 100 percent (that is, once life gets a foothold, intelligent life is sure to follow) to close to 0 percent (that is, intelligent life is very rare). f_c : For each planet with intelligent life, what is the fraction that communicates with radio waves? The estimates for f_c tend to be higher than for f_l and f_i , based on the (sensible) reasoning that once you have an intelligent species, the discovery and use of radio communication is likely. f_L = the fraction of the universe's life during which an average communicating civilization communicates with radio waves.⁶⁹ If we take our civilization as an example, we have been communicating with radio transmissions for about one hundred years out of the roughly ten- to twenty-billion-year history of the universe, so f_L for the Earth is about 10^{-8} so far. If we continue communicating with radio waves for, say, another nine hundred years, the factor would then be 10^{-7} . This factor is affected by a number of considerations. If

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Enrico Fermi asked in the summer of 1950, "Where is everybody?"⁷³ A sufficiently advanced civilization would not be likely to restrict its transmissions to subtle signals on obscure frequencies. Why are all the ETIs so shy? There have been attempts to respond to the so-called Fermi Paradox (which, granted, is a paradox only if one accepts the optimistic parameters that most observers apply to the Drake equation).

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In chapter 3 I discussed the ultimate cold laptop and estimated the optimal computational capacity of a one-liter, one-kilogram computer at around 1042 cps, which is sufficient to perform the equivalent of ten thousand years of the thinking of ten billion human brains in ten microseconds.

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since civilization moves rapidly toward nonbiological intelligence once it discovers computation,

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Note that the computational powers of these solar system-scale computers will be achieved, according to my projections in chapter 2, around the end of this century.

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Recent experiments have measured the flight time of photons at nearly twice the speed of light, a result of quantum uncertainty on their position.⁸⁰ However, this result is really not useful for this analysis, because it

does not actually allow information to be communicated faster than the speed of light, and we are fundamentally interested in communication speed.

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Dr. Nicolas Gisin of the University of Geneva sent quantum-entangled photons in opposite directions through optical fibers across Geneva. When the photons were seven miles apart, they each encountered a glass plate. Each photon had to “decide” whether to pass through or bounce off the plate (which previous experiments with non-quantum-entangled photons have shown to be a random choice). Yet because the two photons were quantum entangled, they made the same decision at the same moment. Many repetitions provided the identical result.⁸¹ The experiments have not absolutely ruled out the explanation of a hidden variable—that is, an unmeasurable state of each particle that is in phase (set to the same point in a cycle), so that when one particle is measured (for example, has to decide its path through or off a glass plate), the other has the same value of this internal variable. So the “choice” is generated by an identical setting of this hidden variable, rather than being the result of actual communication between the two particles. However, most quantum physicists reject this interpretation.

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Yet even if we accept the interpretation of these experiments as indicating a quantum link between the two particles, the apparent communication is transmitting only randomness (profound quantum randomness) at speeds far greater than the speed of light, not predetermined information, such as the bits in a file. This communication of quantum random decisions to different points in space could have value, however, in applications such as providing encryption codes. Two different locations could receive the same random sequence, which could then be used by one location to encrypt a message and by the other to decipher it. It would not be possible for anyone else to eavesdrop on the encryption code without destroying the quantum entanglement and thereby being detected. There are already commercial encryption products incorporating this principle. This is a fortuitous application of quantum mechanics because of the possibility that another application of quantum mechanics—quantum computing—may put an end to the standard method of encryption based on factoring large numbers (which quantum computing, with a large number of entangled qubits, would be good at).

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phenomenon is the speed with which galaxies can recede from each other as a result of the expansion of the universe. If the distance between two galaxies is greater than what is called the Hubble distance, then these galaxies are receding from one another at faster than the speed of light.⁸² This does not violate Einstein’s special theory of relativity, because this velocity is caused by space itself expanding rather than the galaxies moving through space. However, it also doesn’t help us transmit information at speeds faster than the speed of light.

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Wormholes. There are two exploratory conjectures that suggest ways to circumvent the apparent limitation of the speed of light. The first is to use wormholes—folds of the universe in dimensions beyond the three visible ones.

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In 1955 physicist John Wheeler described these tunnels as “wormholes,” introducing the term for the first time. 84 His analysis of worm-holes showed them to be fully consistent with the theory of general relativity, which describes space as essentially curved in another dimension.

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Thorne and his Ph.D. students Morris and Yurtsever also described a method consistent with general relativity and quantum mechanics that could establish wormholes between the Earth and faraway locations. Their proposed technique involves expanding a spontaneously generated, subatomic-size worm-hole to a larger size by adding energy, then stabilizing it using superconducting spheres in the two connected “wormhole mouths.” After the wormhole is expanded and stabilized, one of its mouths (entrances) is transported to another location, while keeping its connection to the other entrance, which remains on Earth.

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Physicist David Hochberg and Vanderbilt University’s Thomas Kephart point out that shortly after the Big Bang, gravity was strong enough to have provided the energy required to spontaneously create massive numbers of self-stabilizing wormholes.⁸⁸ A significant portion of these wormholes is likely to still be around and may be pervasive, providing a vast network of corridors that reach far and wide throughout the universe. It might be easier to discover and use these natural wormholes than to create new ones.

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In 2001 astronomer John Webb discovered that the so-called fine-structure constant varied when he examined light from sixty-eight quasars (very bright young galaxies).⁸⁹ The speed of light is one of four constants that the fine-structure constant comprises, so the result is another suggestion that varying conditions in the universe may cause the speed of light to change. Cambridge University physicist John Barrow and his colleagues are in the process of running a two-year tabletop experiment that will test the ability to engineer a small change in the speed of light.⁹⁰ Suggestions that the speed of light can vary are consistent with recent theories that it was significantly higher during the inflationary period of the universe (an early phase in its history, when it underwent very rapid expansion).

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if confirmed, the findings would be profound, because it is the role of engineering to take a subtle effect and greatly amplify it.

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the mental experiment we should perform now is not whether contemporary human scientists, such as we are, can perform these engineering feats but whether or not a human civilization that has expanded its intelligence by trillions of trillions will be able to do so.

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For now we can say that ultrahigh levels of intelligence will expand outward at the speed of light, while recognizing that our contemporary understanding of physics suggests that this may not be the actual limit of the speed of expansion or, even if the speed of light proves to be immutable, that this limit may not restrict reaching other locations quickly through wormholes.

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If they exist, why would they come here? One mission would be for observation—to gather knowledge (just as we observe other species on Earth today). Another would be to seek matter and energy to provide additional substrate for its expanding intelligence. The intelligence and equipment needed for such exploration and expansion (by an ETI, or by us when we get to that stage of development) would be extremely small, basically nanobots and information transmissions.

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Let's first consider the anthropic principle as applied to the universe in more detail. The question concerning the universe arises because we notice that the constants in nature are precisely what are required for the universe to have grown in complexity. If the cosmological constant, the Planck constant, and the many other constants of physics were set to just slightly different values, atoms, molecules, stars, planets, organisms, and humans would have been impossible. The universe appears to have exactly the right rules and constants.

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although we can fashion all kinds of reasons why each particular intelligent civilization may remain hidden from us (for example, they destroyed themselves, or they have decided to remain invisible or stealthy, or they've switched all of their communications away from electromagnetic transmissions, and so on), it is not credible to believe that every single civilization out of the billions that should be there (according to the SETI assumption) has some reason to be invisible.

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the computational power of a computer is a function of its mass and its computational efficiency.

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Once we achieve an optimal computational efficiency, the only way to increase the computational power of a computer would be to increase its mass. If we increase the mass enough, its gravitational force becomes strong enough to cause it to collapse into a black hole. So a black hole can be regarded as the ultimate computer. Of course, not any black hole will do. Most black holes, like most rocks, are performing lots of random transactions but no useful computation. But a well-organized black hole would be the most powerful conceivable computer in terms of cps per liter.

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on July 21, 2004, Hawking admitted defeat and acknowledged that Preskill had been correct after all: that information sent into a black hole is not lost. It could be transformed inside the black hole and then transmitted outside it. According to this understanding, what happens is that the particle that flies away from the black hole remains quantum entangled with its antiparticle that disappeared into the black hole. If that antiparticle inside the black hole becomes involved in a useful computation, then these results will be encoded in the state of its tangled partner particle outside of the black hole.

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Assuming that Hawking's new position is indeed correct, the ultimate computers that we can create would be black holes. Therefore a universe that is well designed to create black holes would be one that is well designed to optimize its intelligence. Susskind and Smolin argued merely that biology and black holes both require the same kind of materials, so a universe that was optimized for black holes would also be optimized for biology. Recognizing that black holes are the ultimate repository of intelligent computation, however, we can conclude that the utility function of optimizing black-hole production and that of optimizing intelligence are one and the same.

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another reason to apply an anthropic principle.

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Although smart matter still nominally follows the laws of physics, it is so extraordinarily intelligent that it can harness the most subtle aspects of the laws to manipulate matter and energy to its will. So it would at least appear that intelligence is more powerful than physics.

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once matter evolves into smart matter (matter fully saturated with intelligent processes), it can manipulate other matter and energy to do its bidding (through suitably powerful engineering).

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Once a planet yields a technology-creating species and that species creates computation (as has happened here), it is only a matter of a few centuries before its intelligence saturates the matter and energy in its vicinity, and it begins to expand outward at at least the speed of light (with some suggestions of circumventing this limit). Such a civilization will then overcome gravity (through exquisite and vast technology) and other cosmological forces—or, to be fully accurate, it will maneuver and control these forces—and engineer the universe it wants. This is the goal of the Singularity.

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The universe is about 1017 seconds old,

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The Holographic Universe. Another perspective on the maximum information storage and processing capability of the universe comes from a speculative recent theory of the nature of information. According to the “holographic universe” theory the universe is actually a two-dimensional array of information written on its surface, so its conventional three-dimensional appearance is an illusion.⁹⁹ In essence, the universe, according to this theory, is a giant hologram. The information is written at a very fine scale, governed by the Planck constant. So the maximum amount of information in the universe is its surface area divided by the square of the Planck constant, which comes to about 10¹²⁰ bits. There does not appear to be enough matter in the universe to encode this much information, so the limits of the holographic universe may be higher than what is actually feasible. In any event the order of magnitude of the number of orders of magnitudes of these various estimates is in the same range. The number of bits that a universe reorganized for useful computation will be able to store is 10 raised to a power somewhere between 80 and 120. Again, our engineering, even that of our vastly evolved future selves, will probably fall short of these maximums.

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If I had to place a bet, I would put my money on the conjecture that circumventing the speed of light is possible and that we will be able to do this within the next couple of hundred years. But that is speculation on my part, as we do not yet understand these issues sufficiently to make a more definitive statement.

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This large difference in timespans—hundreds of years versus billions of years (to saturate the universe with our intelligence)—demonstrates why the issue of circumventing the speed of light will become so important. It will become a primary preoccupation of the vast intelligence of our civilization in the twenty-second century. That is why I believe that if wormholes or other circumventing means are feasible, we will be highly motivated to find and exploit them.

Highlight (Yellow) | Location 6979

Epoch Six [the stage at which the nonbiological portion of our intelligence spreads through the universe],

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MOLLY 2004: I’m still trying to envision what the universe will be doing in Epoch Six. TIMOTHY LEARY: The universe will be flying like a bird. MOLLY 2004: But what is it flying in? I mean it’s everything. TIMOTHY: That’s like asking, What is the sound of one hand clapping? MOLLY 2004: Hmmm, so the Singularity is what the Zen masters had in mind all along.

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Substrate is morally irrelevant, assuming it doesn't affect functionality or consciousness. It doesn't matter, from a moral point of view, whether somebody runs on silicon or biological neurons (just as it doesn't matter whether you have dark or pale skin). On the same grounds, that we reject racism and speciesism, we should also reject carbon-chauvinism, or bioism. —NICK BOSTROM, "ETHICS FOR INTELLIGENT MACHINES: A PROPOSAL, 2001"

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A Singularitarian is someone who understands the Singularity and has reflected on its meaning for his or her own life.

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George Gilder has described my scientific and philosophical views as "a substitute vision for those who have lost faith in the traditional object of religious belief."1 Gilder's statement is understandable, as there are at least apparent similarities between anticipation of the Singularity and anticipation of the transformations articulated by traditional religions. But I did not come to my perspective as a result of searching for an alternative to customary faith. The origin of my quest to understand technology trends was practical: an attempt to time my inventions and to make optimal tactical decisions in launching technology enterprises. Over time this modeling of technology took on a life of its own and led me to formulate a theory of technology evolution. It was not a huge leap from there to reflect on the impact of these crucial changes on social and cultural institutions and on my own life. So, while being a Singularitarian is not a matter of faith but one of understanding, pondering the scientific trends I've discussed in this book inescapably engenders new perspectives on the issues that traditional religions have attempted to address: the nature of mortality and immortality, the purpose of our lives, and intelligence in the universe.

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how do we contemplate the Singularity? As with the sun, it's hard to look at directly; it's better to squint at it out of the corners of our eyes. As Max More states, the last thing we need is another dogma, nor do we need another cult, so Singularitarianism is not a system of beliefs or unified viewpoints. While it is fundamentally an understanding of basic technology trends, it is simultaneously an insight that causes one to rethink everything, from the nature of health and wealth to the nature of death and self.

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strive to improve these patterns by optimizing the health of our bodies and extending the reach of our minds. Ultimately, we will be able to vastly expand our mental faculties by merging with our technology.

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We need a body, but once we incorporate MNT fabrication into ourselves, we will be able to change our bodies at will.

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In my view the purpose of life—and of our lives—is to create and appreciate ever-greater knowledge, to move toward greater “order.” As I discussed in chapter 2, increasing order usually means increasing complexity, but sometimes a profound insight will increase order while reducing complexity.

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Having reached a tipping point, we will within this century be ready to infuse our solar system with our intelligence through self-replicating nonbiological intelligence. It will then spread out to the rest of the universe.

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Ideas are the embodiment and the product of intelligence. The ideas exist to solve most any problem that we encounter. The primary problems we cannot solve are ones that we cannot articulate and are mostly ones of which we are not yet aware. For the problems that we do encounter, the key challenge is to express them precisely in words (and sometimes in equations). Having done that, we have the ability to find the ideas to confront and resolve each such problem.

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We can apply the enormous leverage provided by the acceleration of technology. A notable example is achieving radical life extension through “a bridge to a bridge to a bridge” (applying today’s knowledge as a bridge to biotechnology, which in turn will bridge us to the era of nanotechnology).⁴ This offers a way to live indefinitely now, even though we don’t yet have all the knowledge necessary for radical life extension. In other words we don’t have to solve every problem today.

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Nietzsche’s observation “Man is a rope, fastened between animal and overman—a rope over an abyss.” We can interpret Nietzsche to be pointing out that we have advanced beyond other animals while seeking to become something far greater. We might regard Nietzsche’s reference to the abyss to allude to the perils inherent in technology,

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More has at the same time expressed concern that anticipating the Singularity could engender a passivity in addressing today’s issues.⁶ Because

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Still Human? Some observers refer to the post-Singularity period as “posthuman” and refer to the anticipation of this period as posthumanism. However, to me being human means being part of a civilization that seeks to extend its boundaries. We are already reaching beyond our biology by rapidly gaining the tools to reprogram and

augment it. If we regard a human modified with technology as no longer human, where would we draw the defining line? Is a human with a bionic heart still human? How about someone with a neurological implant? What about two neurological implants? How about someone with ten nanobots in his brain? How about 500 million nanobots? Should we establish a boundary at 650 million nanobots: under that, you're still human and over that, you're posthuman? Our merger with our technology has aspects of a slippery slope, but one that slides up toward greater promise, not down into Nietzsche's abyss. Some observers refer to this merger as creating a new "species." But the whole idea of a species is a biological concept, and what we are doing is transcending biology. The transformation underlying the Singularity is not just another in a long line of steps in biological evolution. We are upending biological evolution altogether. BILL GATES: I agree with you 99 percent. What I like about your ideas is that they are grounded in science, but your optimism is almost a religious faith. I'm optimistic also.

Highlight (Yellow) | Location 7130

RAY: Yes, well, we need a new religion. A principal role of religion has been to rationalize death, since up until just now there was little else constructive we could do about it. BILL: What would the principles of the new religion be? RAY: We'd want to keep two principles: one from traditional religion and one from secular arts and sciences—from traditional religion, the respect for human consciousness. BILL: Ah yes, the Golden Rule. RAY: Right, our morality and legal system are based on respect for the consciousness of others. If I hurt another person, that's considered immoral, and probably illegal, because I have caused suffering to another conscious person. If I destroy property, it's generally okay if it's my property, and the primary reason it's immoral and illegal if it's someone else's property is because I have caused suffering not to the property but to the person owning it. BILL: And the secular principle? RAY: From the arts and sciences, it is the importance of knowledge. Knowledge goes beyond information. It's information that has meaning for conscious entities: music, art, literature, science, technology. These are the qualities that will expand from the trends I'm talking about. BILL: We need to get away from the ornate and strange stories in contemporary religions and concentrate on some simple messages. We need a charismatic leader for this new religion. RAY: A charismatic leader is part of the old model. That's something we want to get away from. BILL: Okay, a charismatic computer, then. RAY: How about a charismatic operating system? BILL: Ha, we've already got that. So is there a God in this religion? RAY: Not yet, but there will be. Once we saturate the matter and energy in the universe with intelligence, it will "wake up," be conscious, and sublimely intelligent. That's about as close to God as I can imagine.

Highlight (Yellow) | Location 7171

The Vexing Question of Consciousness If you could blow the brain up to the size of a mill and walk about inside, you would not find consciousness. —G. W. LEIBNIZ Can one ever remember love? It's like trying to summon up the smell of roses in a cellar. You might see a rose, but never the perfume. —ARTHUR MILLER⁸ At one's first and simplest attempts to philosophize, one becomes entangled in questions of whether when one knows something, one knows that one knows it, and what, when one is thinking of oneself, is being thought about, and what is doing the thinking. After one has been puzzled and bruised by this problem for a long time, one learns not to press these questions: the concept of a conscious being is, implicitly, realized to be different from that of an unconscious object. In saying that a conscious being knows something, we are saying not only that he knows it, but that he knows that he knows it, and that he knows that he knows that he knows it, and so on, as long as we care to pose the question: there is, we recognize, an infinity here, but it is not an infinite

regress in the bad sense, for it is the questions that peter out, as being pointless, rather than the answers. —J. R. LUCAS, OXFORD PHILOSOPHER, IN HIS 1961 ESSAY “MINDS, MACHINES, AND GÖDEL” “9 Dreams are real while they last; can we say more of life? —HAVELOCK ELLIS

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By the late 2020s we will have completed the reverse engineering of the human brain, which will enable us to create nonbiological systems that match and exceed the complexity and subtlety of humans, including our emotional intelligence. A second scenario is that we could upload the patterns of an actual human into a suitable nonbiological, thinking substrate. A third, and the most compelling, scenario involves the gradual but inexorable progression of humans themselves from biological to nonbiological. That has already started with the benign introduction of devices such as neural implants to ameliorate disabilities and disease. It will progress with the introduction of nanobots in the bloodstream, which will be developed initially for medical and antiaging applications. Later more sophisticated nanobots will interface with our biological neurons to augment our senses, provide virtual and augmented reality from within the nervous system, assist our memories, and provide other routine cognitive tasks. We will then be cyborgs, and from that foothold in our brains, the nonbiological portion of our intelligence will expand its powers exponentially.

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ongoing exponential growth of every aspect of information technology, including price-performance, capacity, and rate of adoption. Given that the mass and energy required to compute and communicate each bit of information are extremely small (see chapter 3), these trends can continue until our nonbiological intelligence vastly exceeds that of the biological portion.

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In the 2040s, when the nonbiological portion will be billions of times more capable, will we still link our consciousness to the biological portion of our intelligence?

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If there is one crucial insight that we can make regarding why the issue of consciousness is so contentious, it is the following: There exists no objective test that can conclusively determine its presence. Science is about objective measurements and their logical implications, but the very nature of objectivity is that you cannot measure subjective experience—you can only measure correlates of it, such as behavior

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There is no consensus among humans about the consciousness of non-human entities, such as higher animals. Consider the debates regarding animal rights, which have everything to do with whether animals are conscious or just quasi machines that operate by “instinct.” The issue will be even more contentious with regard to future nonbiological entities that exhibit behavior and intelligence even more humanlike than those of animals.

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We marvel at Einstein's ability to conjure up the theory of general relativity from a thought experiment or Beethoven's ability to imagine symphonies that he could never hear. But these instances of human thought at its best are rare and fleeting. (Fortunately we have a record of these fleeting moments,

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The issue of who or what is conscious and the nature of the subjective experiences of others are fundamental to our concepts of ethics, morality, and law. Our legal system is based largely on the concept of consciousness, with particularly serious attention paid to actions that cause suffering—an especially acute form of conscious experience—to a (conscious) human or that end the conscious experience of a human (for example, murder).

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there is no device or system we can postulate that could definitively detect subjectivity (conscious experience) associated with an entity. Any such purported device would have philosophical assumptions built into it.

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MOLLY 2104: We experience that very directly in 2104. Being nonbiological, I'm able to change who I am quite readily. As we discussed earlier, if I'm in the mood, I can combine my thought patterns with someone else's and create a merged identity. It's a profound experience. MOLLY 2004: Well, Miss Molly of the future, we do that back in the primitive days of 2004 also. We call it falling in love.

Highlight (Yellow) | Location 7312

What you are looking for is who is looking. —SAINT FRANCIS OF ASSISI

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The specific set of particles that my body and brain comprise are in fact completely different from the atoms and molecules that I comprised only a short while ago. We know that most of our cells are turned over in a matter of weeks, and even our neurons, which persist as distinct cells for a relatively long time, nonetheless change all of their constituent molecules within a month.¹⁴ The half-life of a microtubule (a protein filament that provides the structure of a neuron) is about ten minutes. The actin filaments in dendrites are replaced about every forty seconds. The proteins that power the synapses are replaced about every hour. NMDA receptors in synapses stick around for a relatively long five days. So I am a completely different set of stuff than I was a month ago, and all that persists is the pattern of organization of that stuff.

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Although he would have all my memories and recall having been me, from the point in time of his creation Ray 2 would have his own unique experiences, and his reality would begin to diverge from mine.

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The gradual replacement of Ray results in Ray, so consciousness and identity appear to have been preserved. However, in the case of gradual replacement there is no simultaneous old me and new me. At the end of the process you have the equivalent of the new me (that is, Ray 2) and no old me (Ray 1). So gradual replacement also means the end of me. We might therefore wonder: at what point did my body and brain become someone else?

Highlight (Yellow) | Location 7369

all that persists is my spatial and temporal pattern of matter and energy.

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So, again, who am I? It's the ultimate ontological question, and we often refer to it as the issue of consciousness.

Highlight (Yellow) | Location 7396

Biological humans have about ten trillion cells with their own DNA, but there are about one hundred trillion microorganisms in the digestive tract, basically bacteria. MOLLY 2004: Doesn't sound very appealing. Are they entirely necessary? RAY: They're actually part of the society of cells that makes Molly alive and thriving. You couldn't survive without healthy gut bacteria. Assuming your intestinal flora are in good balance, they're necessary for your well-being. MOLLY 2004: Okay, but I wouldn't count them as me.

Highlight (Yellow) | Location 7416

MOLLY 2004: The boundary of myself is seeming less and less clear. RAY: Well, just wait until we're predominantly nonbiological. Then we'll be able to merge our thoughts and thinking at will, so finding boundaries will be even more difficult. MOLLY 2004: That actually sounds kind of appealing. You know, some Buddhist philosophies emphasize the extent to which there is inherently no boundary at all between us. RAY: Sounds like they're talking about the Singularity.

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As the anthropologist Marshall Sahlins puts the matter: "We are the only people who assume that we have ascended from apes. Everybody else takes it for granted that they are descended from gods." —HUSTON SMITH16

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Or consider Einstein's description of the entangled order and disorder in a glass of water (that is, his thesis on Brownian motion).

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I regard myself as a “patternist.” It’s through the emergent powers of the pattern that we transcend. Since the material stuff of which we are made turns over quickly, it is the transcendent power of our patterns that persists.

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The power of patterns to endure goes beyond explicitly self-replicating systems, such as organisms and self-replicating technology. It is the persistence and power of patterns that support life and intelligence. The pattern is far more important than the material stuff that constitutes it.

Highlight (Yellow) | Location 7447

Random strokes on a canvas are just paint. But when arranged in just the right way, they transcend the material stuff and become art. Random notes are just sounds. Sequenced in an “inspired” way, we have music. A pile of components is just an inventory. Ordered in an innovative manner, and perhaps with the addition of some software (another pattern), we have the “magic” (transcendence) of technology. Although some regard what is referred to as “spiritual” as the true meaning of transcendence, transcendence refers to all levels of reality: the creations of the natural world, including ourselves, as well as our own creations in the form of art, culture, technology, and emotional and spiritual expression.

Highlight (Yellow) | Location 7453

As a consummation of the evolution in our midst, the Singularity will deepen all of these manifestations of transcendence.

Highlight (Yellow) | Location 7454

Consciousness—the seat of “personalness”—is regarded as what is real in many philosophical and religious traditions. A common Buddhist ontology considers subjective—conscious—experience as the ultimate reality, rather than physical or objective phenomena, which are considered maya (illusion).

Highlight (Yellow) | Location 7457

The arguments I make in this book with regard to consciousness are for the purpose of illustrating this vexing and paradoxical (and, therefore, profound) nature of consciousness: how one set of assumptions (that is, that a copy of my mind file either shares or does not share my consciousness) leads ultimately to an opposite view, and vice versa.

Highlight (Yellow) | Location 7499

RAY: The universe is not conscious—yet. But it will be. Strictly speaking, we should say that very little of it is conscious today. But that will change and soon. I expect that the universe will become sublimely intelligent and will wake up in Epoch Six. The only belief I am positing here is that the universe exists. If we make that leap of

faith, the expectation that it will wake up is not so much a belief as an informed understanding, based on the same science that says there is a universe. MOLLY 2004: Interesting. You know, that's essentially the opposite of the view that there was a conscious creator who got everything started and then kind of bowed out. You're basically saying that a conscious universe will "bow in" during Epoch Six. RAY: Yes, that's the essence of Epoch Six.

Highlight (Yellow) | Location 7515

Progress might have been all right once, but it has gone on too long. —OGDEN NASH (1902–1971)

Highlight (Yellow) | Location 7525

... flight from and hatred of technology is self-defeating. The Buddha rests quite as comfortably in the circuits of a digital computer and the gears of a cycle transmission as he does at the top of a mountain or in the petals of a flower. To think otherwise is to demean the Buddha—which is to demean oneself. —ROBERT M. PIRSIG, ZEN AND THE ART OF MOTORCYCLE MAINTENANCE

Highlight (Yellow) | Location 7551

The Web is an extraordinary research tool. In my own experience, research that used to require a half day at the library can now be accomplished typically in a couple of minutes or less. This has enormous and obvious benefits for advancing beneficial technologies, but it can also empower those whose values are inimical to the mainstream of society. So are we in danger? The answer is clearly yes. How much danger, and what to do about it, are the subjects of this chapter.

Highlight (Yellow) | Location 7576

genetics, nanotechnology, and robotics (GNR, as discussed earlier).

Highlight (Yellow) | Location 7593

He suggests, for example, a prohibition against self-replicating nanotechnology, which is similar to the guidelines advocated by the Foresight Institute, founded by nanotechnology pioneer Eric Drexler and Christine Peterson. Overall, this is a reasonable guideline, although I believe there will need to be two exceptions, which I discuss below (see p. 411). As another example, Joy advocates not publishing the gene sequences of pathogens on the Internet, which I also agree with.

Highlight (Yellow) | Location 7599

Others, such as Bill McKibben, the environmentalist who was one of the first to warn against global warming, have advocated relinquishment of broad areas such as biotechnology and nanotechnology, or even of all technology. As I discuss in greater detail below (see p. 410), relinquishing broad fields would be impossible to achieve without essentially relinquishing all technical development. That in turn would require a Brave New World style of totalitarian government, banning all technology development. Not only would such a solution be

inconsistent with our democratic values, but it would actually make the dangers worse by driving the technology underground, where only the least responsible practitioners (for example, rogue states) would have most of the expertise.

Highlight (Yellow) | Location 7606

It was the best of times, it was the worst of times, it was the age of wisdom, it was the age of foolishness, it was the epoch of belief, it was the epoch of incredulity, it was the season of Light, it was the season of Darkness, it was the spring of hope, it was the winter of despair, we had everything before us, we had nothing before us, we were all going direct to Heaven, we were all going direct the other way. —CHARLES DICKENS, A TALE OF TWO CITIES

Highlight (Yellow) | Location 7611

It's like arguing in favor of the plough. You know some people are going to argue against it, but you also know it's going to exist. —JAMES HUGHES, SECRETARY OF THE TRANSHUMANIST ASSOCIATION AND SOCIOLOGIST AT TRINITY COLLEGE, IN A DEBATE, "SHOULD HUMANS WELCOME OR RESIST BECOMING POSTHUMAN?"

Highlight (Yellow) | Location 7625

We've gone from about twenty democracies in the world after World War II to more than one hundred today largely through the influence of decentralized electronic communication. The biggest wave of democratization, including the fall of the Iron Curtain, occurred during the 1990s with the growth of the Internet and related technologies.

Highlight (Yellow) | Location 7628

Ubiquitous N and R are two to three decades away and will continue an exponential expansion of these benefits. As I reviewed in earlier chapters, these technologies will create extraordinary wealth, thereby overcoming poverty and enabling us to provide for all of our material needs by transforming inexpensive raw materials and information into any type of product.

Highlight (Yellow) | Location 7631

We will spend increasing portions of our time in virtual environments and will be able to have any type of desired experience with anyone, real or simulated, in virtual reality. Nanotechnology will bring a similar ability to morph the physical world to our needs and desires. Lingering problems from our waning industrial age will be overcome. We will be able to reverse remaining environmental destruction. Nanoengineered fuel cells and solar cells will provide clean energy. Nanobots in our physical bodies will destroy pathogens, remove debris such as misformed proteins and protofibrils, repair DNA, and reverse aging. We will be able to redesign all of the systems in our bodies and brains to be far more capable and durable. Most significant will be the merger of biological and nonbiological intelligence, although nonbiological intelligence will quickly come to predominate. There will be a vast expansion of the concept of what it means to be human. We will greatly enhance our ability

to create and appreciate all forms of knowledge from science to the arts, while extending our ability to relate to our environment and one another. On the other hand and Dangers

Highlight (Yellow) | Location 7646

the twentieth century saw technology's awesome ability to amplify our destructive nature, from Stalin's tanks to Hitler's trains. The tragic event of September 11, 2001, is another example of technologies (jets and buildings) taken over by people with agendas of destruction. We still live today with a sufficient number of nuclear weapons (not all of which are accounted for) to end all mammalian life on the planet.

Highlight (Yellow) | Location 7649

Since the 1980s the means and knowledge have existed in a routine college bioengineering lab to create unfriendly pathogens potentially more dangerous than nuclear weapons.⁹ In a war-game simulation conducted at Johns Hopkins University called "Dark Winter," it was estimated that an intentional introduction of conventional smallpox in three U.S. cities could result in one million deaths. If the virus were bioengineered to defeat the existing smallpox vaccine, the results could be far worse.

Highlight (Yellow) | Location 7661

Although the possibility of an even more virulent return of SARS remains uncertain as of the writing of this book, it appears that containment measures have been relatively successful and have prevented this tragic outbreak from becoming a true catastrophe. Part of the response involved ancient, low-tech tools such as quarantine and face masks.

Highlight (Yellow) | Location 7664

this approach would not have worked without advanced tools that have only recently become available. Researchers were able to sequence the DNA of the SARS virus within thirty-one days of the outbreak—compared to fifteen years for HIV. That enabled the rapid development of an effective test so that carriers could quickly be identified.

Highlight (Yellow) | Location 7666

instantaneous global communication facilitated a coordinated response worldwide, a feat not possible when viruses ravaged the world in ancient times.

Highlight (Yellow) | Location 7669

A quintessential concern that has received considerable attention is unrestrained nanobot replication. Nanobot technology requires trillions of such intelligently designed devices to be useful.

Highlight (Yellow) | Location 7670

To scale up to such levels it will be necessary to enable them to self-replicate, essentially the same approach used in the biological world (that's how one fertilized egg cell becomes the trillions of cells in a human). And in the same way that biological self-replication gone awry (that is, cancer) results in biological destruction, a defect in the mechanism curtailing nanobot self-replication—the so-called gray-goo scenario—would endanger all physical entities, biological or otherwise.

Highlight (Yellow) | Location 7674

Living creatures—including humans—would be the primary victims of an exponentially spreading nanobot attack. The principal designs for nanobot construction use carbon as a primary building block. Because of carbon's unique ability to form four-way bonds, it is an ideal building block for molecular assemblies. Carbon molecules can form straight chains, zigzags, rings, nanotubes (hexagonal arrays formed in tubes), sheets, buckyballs (arrays of hexagons and pentagons formed into spheres), and a variety of other shapes. Because biology has made the same use of carbon, pathological nanobots would find the Earth's biomass an ideal source of this primary ingredient. Biological entities can also provide stored energy in the form of glucose and ATP.¹³

Highlight (Yellow) | Location 7680

How long would it take an out-of-control replicating nanobot to destroy the Earth's biomass? The biomass has on the order of 10⁴⁵ carbon atoms.

Highlight (Yellow) | Location 7689

more insidious possibility. In a two-phased attack, the nanobots take several weeks to spread throughout the biomass but use up an insignificant portion of the carbon atoms, say one out of every thousand trillion (10¹⁵). At this extremely low level of concentration the nanobots would be as stealthy as possible. Then, at an "optimal" point, the second phase would begin with the seed nanobots expanding rapidly in place to destroy the biomass. For each seed nanobot to multiply itself a thousand trillionfold would require only about fifty binary replications, or about ninety minutes.

Highlight (Yellow) | Location 7694

The point is that without defenses, the available biomass could be destroyed by gray goo very rapidly.

Highlight (Yellow) | Location 7701

The nanotechnology immune system mentioned above, for example, will ultimately require self-replication; otherwise it would be unable to defend us.

Highlight (Yellow) | Location 7708

basic elements available in airborne dust and sunlight for power.

Highlight (Yellow) | Location 7709

using carbon and other elements on rocks.

Highlight (Yellow) | Location 7716

Biological life on Earth encountered a human-made existential risk for the first time in the middle of the twentieth century with the advent of the hydrogen bomb and the subsequent cold-war buildup of thermonuclear forces. President Kennedy reportedly estimated that the likelihood of an all-out nuclear war during the Cuban missile crisis was between 33 and 50 percent.¹⁹

Highlight (Yellow) | Location 7732

Some viruses are easily communicable, such as the flu and common cold. Others are deadly, such as HIV. It is rare for a virus to combine both attributes.

Highlight (Yellow) | Location 7736

Other viruses, such as smallpox, have both negative characteristics—they are easily contagious and deadly—but have been around long enough that there has been time for society to create a technological protection in the form of a vaccine. Gene engineering, however, has the potential to bypass these evolutionary protections by suddenly introducing new pathogens for which we have no protection, natural or technological. The prospect of adding genes for deadly toxins to easily transmitted, common viruses such as the common cold and flu introduced another possible existential-risk scenario.

Highlight (Yellow) | Location 7742

The emergence of SARS resulted from a combination of an ancient practice (the virus is suspected of having jumped from exotic animals, possibly civet cats, to humans living in close proximity) and a modern practice (the infection spread rapidly across the world by air travel). SARS provided us with a dry run of a virus new to human civilization that combined easy transmission, the ability to survive for extended periods of time outside the human body, and a high degree of mortality, with death rates estimated at 14 to 20 percent. Again, the response combined ancient and modern techniques.

Highlight (Yellow) | Location 7754

the window of malicious opportunity for bioengineered viruses, existential or otherwise, will close in the 2020s when we have fully effective antiviral technologies based on nanobots.²³ However, because nanotechnology will be thousands of times stronger, faster, and more intelligent than biological entities, self-replicating nanobots will present a greater risk and yet another existential risk. The window for malevolent nanobots will ultimately be closed by strong artificial intelligence, but, not surprisingly, “unfriendly” AI will itself present an even more compelling existential risk,

Highlight (Yellow) | Location 7760

what has become known as the “precautionary principle.” (If the consequences of an action are unknown but judged by some scientists to have even a small risk of being profoundly negative, it’s better to not carry out the action than risk negative consequences.)

Highlight (Yellow) | Location 7766

Max More articulates the limitations of the precautionary principle and advocates replacing it with what he calls the “proactionary principle,” which involves balancing the risks of action and inaction.²⁴

Highlight (Yellow) | Location 7769

The Smaller the Interaction, the Larger the Explosive Potential. There has been recent controversy over the potential for future very high-energy particle accelerators to create a chain reaction of transformed energy states at a subatomic level. The result could be an exponentially spreading area of destruction, breaking apart all atoms in our galactic vicinity. A variety of such scenarios has been proposed, including the possibility of creating a black hole that would draw in our solar system. Analyses of these scenarios show them to be very unlikely, although not all physicists are sanguine about the danger.²⁵ The mathematics of these analyses appears to be sound, but we do not yet have a consensus

Highlight (Yellow) | Location 7775

If such dangers sound far-fetched, consider the possibility that we have indeed detected increasingly powerful explosive phenomena at diminishing scales of matter. Alfred Nobel discovered dynamite by probing chemical interactions of molecules. The atomic bomb, which is tens of thousands of times more powerful than dynamite, is based on nuclear interactions involving large atoms, which are much smaller scales of matter than large molecules. The hydrogen bomb, which is thousands of times more powerful than an atomic bomb, is based on interactions involving an even smaller scale: small atoms. Although this insight does not necessarily imply the existence of yet more powerful destructive chain reactions by manipulating subatomic particles, it does make the conjecture plausible.

Highlight (Yellow) | Location 7784

Stumbling across the exact conditions to create a new class of catastrophic chain reaction at a subatomic level appears to be even less likely.

Highlight (Yellow) | Location 7790

The best way we could avoid being shut down would be to be interesting to the observers of the simulation. Assuming that someone is actually paying attention to the simulation, it’s a fair assumption that it’s less likely to be turned off when it’s compelling

Highlight (Yellow) | Location 7795

achieving a Singularity of exploding knowledge may be the very purpose of the simulation. Thus, assuring a “constructive” Singularity (one that avoids degenerate outcomes such as existential destruction by gray goo or dominance by a malicious AI) could be the best course to prevent the simulation from being terminated.

Highlight (Yellow) | Location 7816

I do think we also need to take seriously the misguided and increasingly strident Luddite voices that advocate reliance on broad relinquishment of technological progress to avoid the genuine dangers of GNR.

Highlight (Yellow) | Location 7821

a totalitarian brave new world is unlikely because of the democratizing impact of increasingly powerful decentralized electronic and photonic communication.

Highlight (Yellow) | Location 7822

The advent of worldwide, decentralized communication epitomized by the Internet and cell phones has been a pervasive democratizing force.

Highlight (Yellow) | Location 7824

broke decades of totalitarian control of information.

Highlight (Yellow) | Location 7825

The movement toward democracy and capitalism and the attendant economic growth that characterized the 1990s were all fueled by the accelerating force of these person-to-person communication technologies. There are other questions that are nonexistential but nonetheless serious.

Highlight (Yellow) | Location 7827

“Who is controlling the nanobots?” and “Whom are the nanobots talking to?” Future organizations (whether governments or extremist groups) or just a clever individual could put trillions of undetectable nanobots in the water or food supply of an individual or of an entire population. These spybots could then monitor, influence, and even control thoughts and actions. In addition existing nanobots could be influenced through software viruses and hacking techniques. When there is software running in our bodies and brains (as we discussed, a threshold we have already passed for some people), issues of privacy and security will take on a new urgency, and countersurveillance methods of combating such intrusions will be devised.

Highlight (Yellow) | Location 7836

The key features for a wide range of applications, including electronics, mechanics, energy, and medicine, are shrinking at the rate of a factor of about four per linear dimension per decade.

Highlight (Yellow) | Location 7843

strong AI (AI at human levels and beyond)

Highlight (Yellow) | Location 7844

Once AI achieves human levels, it will necessarily soar past it because it will combine the strengths of human intelligence with the speed, memory capacity, and knowledge sharing that nonbiological intelligence already exhibits.

Highlight (Yellow) | Location 7845

Unlike biological intelligence, nonbiological intelligence will also benefit from ongoing exponential gains in scale, capacity, and price-performance.

Highlight (Yellow) | Location 7847

Totalitarian Relinquishment. The only conceivable way that the accelerating pace of advancement on all of these fronts could be stopped would be through a worldwide totalitarian system that relinquishes the very idea of progress.

Highlight (Yellow) | Location 7850

such a totalitarian outcome is unlikely because the increasing decentralization of knowledge is inherently a democratizing force.

Highlight (Yellow) | Location 7860

Two hundred years ago life expectancy for females in the record-holding country (Sweden) was roughly thirty-five years, very brief compared to the longest life expectancy today—almost eighty-five years, for Japanese women. Life expectancy for males was roughly thirty-three years, compared to the current seventy-nine years in the record-holding countries.²⁸ It took half the day to prepare the evening meal, and hard labor characterized most human activity.

Highlight (Yellow) | Location 7865

Only technology, with its ability to provide orders of magnitude of improvement in capability and affordability, has the scale to confront problems such as poverty, disease, pollution, and the other overriding concerns of society today.

Highlight (Yellow) | Location 7867

People often go through three stages in considering the impact of future technology: awe and wonderment at its potential to overcome age-old problems; then a sense of dread at a new set of grave dangers that accompany

these novel technologies; followed finally by the realization that the only viable and responsible path is to set a careful course that can realize the benefits while managing the dangers.

Highlight (Yellow) | Location 7872

The “NBC” (nuclear, biological, and chemical) technologies of warfare have all been used or been threatened to be used in our recent past.²⁹ The far more powerful GNR technologies threaten us with new, profound local and existential risks.

Highlight (Yellow) | Location 7886

technological advances, such as antibiotics and improved sanitation, have freed us from the prevalence of such plagues, and such constructive applications, therefore, need to continue.

Highlight (Yellow) | Location 7888

Should we tell the millions of people afflicted with cancer and other devastating conditions that we are canceling the development of all bioengineered treatments because there is a risk that these same technologies may someday be used for malevolent purposes? Having posed this rhetorical question, I realize that there is a movement to do exactly that, but most people would agree that such broad-based relinquishment is not the answer.

Highlight (Yellow) | Location 7893

The ongoing acceleration of many intertwined technologies produces roads paved with gold. (I use the plural here because technology is clearly not a single path.) In a competitive environment it is an economic imperative to go down these roads. Relinquishing technological advancement would be economic suicide for individuals, companies, and nations.

Highlight (Yellow) | Location 7901

at what level are we to relinquish technology? Ted Kaczynski, who became known to the world as the Unabomber, would have us renounce all of it.³¹

Highlight (Yellow) | Location 7906

“One beer is good, two beers may be better; eight beers, you’re almost certainly going to regret.”³² That metaphor misses the point

Highlight (Yellow) | Location 7908

Although new technologies, like anything else, may be used to excess at times, their promise is not just a matter of adding a fourth cell phone or doubling the number of unwanted e-mails. Rather, it means perfecting the technologies to conquer cancer and other devastating diseases, creating ubiquitous wealth to overcome poverty,

cleaning up the environment from the effects of the first industrial revolution (an objective articulated by McKibben), and overcoming many other age-old problems.

Highlight (Yellow) | Location 7934

once strong AI is merged with nanotechnology, and the ecology of nanoengineered entities becomes highly varied and complex, my own expectation is that we will find that the defending nanorobots need the ability to replicate in place quickly. The other exception is the need for self-replicating nanobot-based probes to explore planetary systems outside of our solar system.

Highlight (Yellow) and Note | Location 7943

A nanocomputer would augment or replace the nucleus in every cell and provide the DNA codes. A nanobot that incorporated molecular machinery similar to ribosomes (the molecules that interpret the base pairs in the mRNA outside the nucleus) would take the codes and produce the strings of amino acids. Since we could control the nanocomputer through wireless messages, we would be able to shut off unwanted replication, thereby eliminating cancer.

albeit just as scary from the hacking/virus perspective

Highlight (Yellow) | Location 7955

Many people are not aware that our intelligence agencies already use advanced technologies such as automated keyword spotting to monitor a substantial flow of telephone, cable, satellite, and Internet conversations. As we go forward, balancing our cherished rights of privacy with our need to be protected from the malicious use of powerful twenty-first-century technologies will be one of many profound challenges. This is one reason such issues as an encryption “trapdoor” (in which law-enforcement authorities would have access to otherwise secure information) and the FBI’s Carnivore e-mail-snooping system have been controversial.³⁷

Highlight (Yellow) | Location 7960

As a test case we can take a small measure of comfort from how we have dealt with one recent technological challenge. There exists today a new fully nonbiological self-replicating entity that didn’t exist just a few decades ago: the computer virus. When this form of destructive intruder first appeared, strong concerns were voiced that as they became more sophisticated, software pathogens had the potential to destroy the computer-network medium in which they live. Yet the “immune system” that has evolved in response to this challenge has been largely effective. Although destructive self-replicating software entities do cause damage from time to time, the injury is but a small fraction of the benefit we receive from the computers and communication links that harbor them.

Highlight (Yellow) | Location 7969

The fact that computer viruses are not usually deadly to humans only means that more people are willing to create and release them. The vast majority of software-virus authors would not release viruses if they thought

they would kill people. It also means that our response to the danger is that much less intense. Conversely, when it comes to self-replicating entities that are potentially lethal on a large scale, our response on all levels will be vastly more serious.

Highlight (Yellow) | Location 7973

Keep in mind that our success in combating them has taken place in an industry in which there is no regulation and minimal certification for practitioners. The largely unregulated computer industry is also enormously productive. One could argue that it has contributed more to our technological and economic progress than any other enterprise in human history. But the battle concerning software viruses and the panoply of software pathogens will never end.

Highlight (Yellow) | Location 7977

When we have software running in our brains and bodies and controlling the world's nanobot immune system, the stakes will be immeasurably greater.

Highlight (Yellow) | Location 7979

The world is struggling with an especially pernicious form of religious fundamentalism in the form of radical Islamic terrorism.

Highlight (Yellow) | Location 7980

they do have an agenda that goes beyond literal interpretations of ancient scriptures: essentially, to turn the clock back on such modern ideas as democracy, women's rights, and education.

Highlight (Yellow) | Location 7983

quoted Patrick Moore, cofounder of Greenpeace, on his disillusionment with the movement he helped found. The issue that undermined Moore's support of Greenpeace was its total opposition to Golden Rice, a strain of rice genetically modified to contain high levels of beta-carotene, the precursor to vitamin A.³⁸ Hundreds of millions of people in Africa and Asia lack sufficient vitamin A, with half a million children going blind each year from the deficiency, and millions more contracting other related diseases. About seven ounces a day of Golden Rice would provide 100 percent of a child's vitamin A requirement.

Highlight (Yellow) | Location 7991

It is not my position that all GMOs are inherently safe; obviously safety testing of each product is needed. But the anti-GMO movement takes the position that every GMO is by its very nature hazardous, a view that has no scientific basis.

Highlight (Yellow) | Location 7998

Certain segments of the environmental movement have become fundamentalist Luddites—“fundamentalist” because of their misguided attempt to preserve things as they are (or were); “Luddite” because of the reflexive stance against technological solutions to outstanding problems. Ironically it is GMO plants—many of which are designed to resist insects and other forms of blight and thereby require greatly reduced levels of chemicals, if any—that offer the best hope for reversing environmental assault from chemicals such as pesticides.

Highlight (Yellow) and Note | Location 8003

Ludditism is inherently fundamentalist. It reflects the idea that humanity will be better off without change, without progress. This brings us back to the idea of relinquishment, as the enthusiasm for relinquishing technology on a broad scale is coming from the same intellectual sources and activist groups that make up the Luddite segment of the environmental movement.

unless the luddite is focussed on their participation and not overall relinquishment

Highlight (Yellow) | Location 8010

In the end, it is only technology—especially GNR—that will offer the leverage needed to overcome problems that human civilization has struggled with for many generations.

Highlight (Yellow) | Location 8014

A phenomenon like gray goo (unrestrained nanobot replication) will be countered with “blue goo” (“police” nanobots that combat the “bad” nanobots).

Highlight (Yellow) | Location 8016

the surest way to prevent development of effective defensive technologies would be to relinquish the pursuit of knowledge in a number of broad areas.

Highlight (Yellow) | Location 8017

We have been able to largely control harmful software-virus replication because the requisite knowledge is widely available to responsible practitioners. Attempts to restrict such knowledge would have given rise to a far less stable situation.

Highlight (Yellow) | Location 8022

While a bioterrorist does not need to put his “inventions” through the FDA, we do require the scientists developing defensive technologies to follow existing regulations, which slow down the innovation process at every step.

Highlight (Yellow) | Location 8027

In terms of public policy the task at hand is to rapidly develop the defensive steps needed, which include ethical standards, legal standards, and defensive technologies themselves.

Highlight (Yellow) | Location 8028

in the software field defensive technologies have responded quickly to innovations in the offensive ones. In the medical field, in contrast, extensive regulation slows down innovation, so we cannot have the same confidence with regard to the abuse of biotechnology.

Highlight (Yellow) | Location 8038

We are close to developing more generalized antiviral technologies, such as RNA interference, and these need to be accelerated. We're addressing biotechnology here because that is the immediate threshold and challenge that we now face. As the threshold for self-organizing nanotechnology approaches, we will then need to invest specifically in the development of defensive technologies in that area, including the creation of a technological immune system.

Highlight (Yellow) | Location 8041

When the body detects a pathogen the T cells and other immune-system cells self-replicate rapidly to combat the invader. A nanotechnology immune system would work similarly both in the human body and in the environment and would include nanobot sentinels that could detect rogue self-replicating nanobots. When a threat was detected, defensive nanobots capable of destroying the intruders would rapidly be created (eventually with self-replication) to provide an effective defensive force.

Highlight (Yellow) | Location 8045

observers have pointed out that such an immune system would itself be a danger because of the potential of "autoimmune" reactions (that is, the immune-system nanobots attacking the world they are supposed to defend).
42 However this possibility is not a compelling reason to avoid the creation of an immune system.

Highlight (Yellow) | Location 8049

the development of a technological immune system for nanotechnology will happen even without explicit efforts to create one. This has effectively happened with regard to software viruses, creating an immune system not through a formal grand-design project but rather through incremental responses to each new challenge and by developing heuristic algorithms for early detection. We can expect the same thing will happen as challenges from nanotechnology-based dangers emerge.

Highlight (Yellow) | Location 8056

Foresight Institute, as one example, has devised a set of ethical standards and strategies for assuring the development of safe nanotechnology, based on guidelines for biotechnology.⁴³ When gene-splicing began in

1975 two biologists, Maxine Singer and Paul Berg, suggested a moratorium on the technology until safety concerns could be addressed. It seemed apparent that there was substantial risk if genes for poisons were introduced into pathogens, such as the common cold, that spread easily. After a ten-month moratorium guidelines were agreed to at the Asilomar conference, which included provisions for physical and biological containment, bans on particular types of experiments, and other stipulations. These biotechnology guidelines have been strictly followed, and there have not been reported accidents in the thirty-year history

Highlight (Yellow) | Location 8062

organization representing the world's organ transplantation surgeons has adopted a moratorium on the transplantation of vascularized animal organs into humans. This was done out of fear of the spread of long-dormant HIV-type xenoviruses from animals such as pigs or baboons into the human population. Unfortunately, such a moratorium can also slow down the availability of lifesaving xenografts (genetically modified animal organs that are accepted by the human immune system) to the millions of people who die each year from heart, kidney, and liver disease. Geoethicist Martine Rothblatt has proposed replacing this moratorium with a new set of ethical guidelines and regulations.⁴⁴

Highlight (Yellow) and Note | Location 8067

In the case of nanotechnology, the ethics debate has started a couple of decades prior to the availability of the particularly dangerous applications. The most important provisions of the Foresight Institute guidelines include:

- “Artificial replicators must not be capable of replication in a natural, uncontrolled environment.”
- “Evolution within the context of a self-replicating manufacturing system is discouraged.”
- “MNT device designs should specifically limit proliferation and provide traceability of any replicating systems.”
- “Distribution of molecular manufacturing development capability should be restricted whenever possible, to responsible actors that have agreed to use the Guidelines. No such restriction need apply to end products of the development process.”

Other strategies that the Foresight Institute has proposed include:

- Replication should require materials not found in the natural environment.
- Manufacturing (replication) should be separated from the functionality of end products. Manufacturing devices can create end products but cannot replicate themselves, and end products should have no replication capabilities.
- Replication should require replication codes that are encrypted and time limited. The broadcast architecture mentioned earlier is an example of this recommendation. These guidelines and strategies are likely to be effective for preventing accidental release of dangerous self-replicating nanotechnology entities. But dealing with the intentional design and release of such entities is a more complex and challenging problem.

bad actors will disregard and defeat encryption and drm methods etc

Highlight (Yellow) and Note | Location 8087

build in

needs editing

Highlight (Yellow) and Note | Location 8090

each level of protection will work only to a certain level of sophistication. The meta-lesson here is that we will need to place twenty-first-century society's highest priority on the continuing advance of defensive technologies, keeping them one or more steps ahead of the destructive technologies (or at least no more than a quick step behind).

like av, etc defense in depth + cat/mouse and raising the bar is the pragmatic solution. Not sure how viable the is if strong AI is the bad actor (though if the pragmatist is also strong AI...)

Highlight (Yellow) and Note | Location 8096

Eliezer Yudkowsky has extensively analyzed paradigms, architectures, and ethical rules that may help assure that once strong AI has the means of accessing and modifying its own design it remains friendly to biological humanity and supportive of its values. Given that self-improving strong AI cannot be recalled, Yudkowsky points out that we need to "get it right the first time," and that its initial design must have "zero nonrecoverable errors."⁴⁵

hmmm. good luck...

Highlight (Yellow) and Note | Location 8099

there will be no absolute protection against strong AI. Although the argument is subtle I believe that maintaining an open free-market system for incremental scientific and technological progress, in which each step is subject to market acceptance, will provide the most constructive environment for technology to embody widespread human values.

the open source / crowd sourced case? ...maybe the former provides a tinge of hope?

Highlight (Yellow) | Location 8102

strong AI is emerging from many diverse efforts and will be deeply integrated into our civilization's infrastructure. Indeed, it will be intimately embedded in our bodies and brains. As such, it will reflect our values because it will be us. Attempts to control these technologies via secretive government programs, along with inevitable underground development, would only foster an unstable environment in which the dangerous applications would be likely to become dominant.

Highlight (Yellow) | Location 8107

Centralized technologies involve an aggregation of resources such as people (for example, cities, buildings), energy (such as nuclear-power plants, liquid-natural-gas and oil tankers, energy pipelines), transportation (airplanes, trains), and other items. Centralized technologies are subject to disruption and disaster. They also tend to be inefficient, wasteful, and harmful to the environment. Distributed technologies, on the other hand, tend to be flexible, efficient, and relatively benign in their environmental effects. The quintessential distributed technology is the Internet. The Internet has not been substantially disrupted to date,

Highlight (Yellow) | Location 8116

nanoengineered solar panels will be able to meet our energy needs in a distributed, renewable, and clean fashion.

Highlight (Yellow) | Location 8118

our need for aggregating people in large buildings and cities will diminish, and people will spread out, living where they want and gathering together in virtual reality.

Highlight (Yellow) | Location 8120

Civil Liberties in an Age of Asymmetric Warfare. The nature of terrorist attacks and the philosophies of the organizations behind them highlight how civil liberties can be at odds with legitimate state interests in surveillance and control.

Highlight (Yellow) | Location 8131

How can the authorities possibly break up a vast network of decentralized cells of suicide terrorists if they have to wait for each one to commit a crime?

Highlight (Yellow) | Location 8132

On the other hand this very logic has been routinely used by tyrannical regimes to justify the waiving of the judicial protections we have come to cherish. It is likewise fair to argue that curtailing civil liberties in this way is exactly the aim of the terrorists, who despise our notions of freedoms and pluralism.

Highlight (Yellow) | Location 8141

We come from goldfish, essentially, but that [doesn't] mean we turned around and killed all the goldfish.

Highlight (Yellow) | Location 8142

If you had a machine with a 10 to the 18th power IQ over humans, wouldn't you want it to govern, or at least control your economy? —SETH SHOSTAK

Highlight (Yellow) | Location 8144

How can we secure the profound benefits of GNR while ameliorating its perils?

Highlight (Yellow) | Location 8145

The most urgent recommendation is to greatly increase our investment in defensive technologies. Since we are already in the G era, the bulk of this investment today should be in (biological) antiviral medications and treatments. We have new tools that are well suited to this task. RNA interference, for example, can be used to

block gene expression. Virtually all infections (as well as cancer) rely on gene expression at some point during their life cycles.

Highlight (Yellow) | Location 8149

Efforts to anticipate the defensive technologies needed to safely guide N and R should also be supported, and these should be substantially increased as we get closer to the feasibility of molecular manufacturing and strong AI, respectively.

Highlight (Yellow) | Location 8155

we need to better balance the risks of new technology (for example, new medications) against the known harm of delay.

Highlight (Yellow) | Location 8157

Diagnostic tools exist to rapidly identify the existence of unknown protein or nucleic acid sequences.

Highlight (Yellow) | Location 8158

Such a “pathogen sentinel” program has been proposed for many years by public health authorities but has never received adequate funding.

Highlight (Yellow) | Location 8163

Efforts to define safety and ethical guidelines for nanotechnology should continue. Such guidelines will inevitably become more detailed and refined as we get closer to molecular manufacturing.

Highlight (Yellow) | Location 8180

deal with the danger from pathological R (strong AI). Our primary strategy in this area should be to optimize the likelihood that future nonbiological intelligence will reflect our values of liberty, tolerance, and respect for knowledge and diversity. The best way to accomplish this is to foster those values in our society today and going forward. If this sounds vague, it is.

Highlight (Yellow) | Location 8185

The transbiological phase will involve nonbiological intelligence deeply integrated with biological intelligence. This will amplify our abilities, and our application of these greater intellectual powers will be governed by the values of its creators. The transbiological era will ultimately give way to the post-biological era, but it is to be hoped that our values will remain influential. This strategy is certainly not foolproof, but it is the primary means we have today to influence the future course of strong AI.

Highlight (Yellow) | Location 8226

MOLLY 2004: So a software virus could turn the nanobot immune system into a stealth destroyer? RAY: That's possible. It's fair to conclude that software security is going to be the decisive issue for many levels of the human-machine civilization. With everything becoming information, maintaining the software integrity of our defensive technologies will be critical to our survival. Even on an economic level, maintaining the business model that creates information will be critical to our well-being.

Highlight (Yellow) | Location 8236

Providing for a reliable nanotechnology immune system will be one of the great political issues of the 2020s and 2030s.

Highlight (Yellow) | Location 8268

The "criticism from analog processing": Digital computation is too rigid because digital bits are either on or off. Biological intelligence is mostly analog, so subtle gradations can be considered. It's true that the human brain uses digital-controlled analog methods, but we can also use such methods in our machines. Moreover, digital computation can simulate analog transactions to any desired level of accuracy, whereas the converse statement is not true.

Highlight (Yellow) | Location 8274

We have achieved realistic mathematical models and computer simulations of neurons and interneuronal connections that do capture the nonlinearities and intricacies of their biological counterparts.

Highlight (Yellow) | Location 8348

we find remarkably precise and predictable exponential trends when assessing the overall effectiveness (as measured by price-performance, bandwidth, and other measures of capability) of information technologies. For example, the smooth exponential growth of the price-performance of computing dates back over a century.

Highlight (Yellow) | Location 8357

The history of human thought is marked by many attempts to refuse to accept ideas that seem to threaten the accepted view that our species is special. Copernicus's insight that the Earth was not at the center of the universe was resisted, as was Darwin's that we were only slightly evolved from other primates. The notion that machines could match and even exceed human intelligence appears to challenge human status once again.

Highlight (Yellow) | Location 8361

something essentially special, after all, about human beings. We were the first species on Earth to combine a cognitive function and an effective opposable appendage (the thumb), so we were able to create technology that would extend our own horizons. No other species on Earth has accomplished this.

Highlight (Yellow) | Location 8377

As I discussed in chapter 3, the amount of matter and energy required to compute or transmit one bit is vanishingly small. By using reversible logic gates, the input of energy is required only to transmit results and to correct errors. Otherwise, the heat released from each computation is immediately recycled to fuel the next computation.

Highlight (Yellow) | Location 8379

As I discussed in chapter 5, nanotechnology-based designs for virtually all applications—computation, communication, manufacturing, and transportation—will require substantially less energy than they do today.

Highlight (Yellow) | Location 8381

Nanotechnology will also facilitate capturing renewable energy sources such as sunlight. We could meet all of our projected energy needs of thirty trillion watts in 2030 with solar power if we captured only 0.03 percent (three ten-thousandths) of the sun's energy as it hit the Earth. This will be feasible with extremely inexpensive, lightweight, and efficient nanoengineered solar panels together with nano-fuel cells to store and distribute the captured energy.

Highlight (Yellow) | Location 8384

A Virtually Unlimited Limit. As I discussed in chapter 3 an optimally organized 2.2-pound computer using reversible logic gates has about 10^{25} atoms and can store about 10^{27} bits. Just considering electromagnetic interactions between the particles, there are at least 10^{15} state changes per bit per second that can be harnessed for computation, resulting in about 10^{42} calculations per second in the ultimate “cold” 2.2-pound computer. This is about 10^{16} times more powerful than all biological brains today. If we allow our ultimate computer to get hot, we can increase this further by as much as 108-fold. And we obviously won't restrict our computational resources to one kilogram of matter but will ultimately deploy a significant fraction of the matter and energy on the Earth and in the solar system and then spread out from there.

Highlight (Yellow) | Location 8391

Specific paradigms do reach limits. We expect that Moore's Law (concerning the shrinking of the size of transistors on a flat integrated circuit) will hit a limit over the next two decades. The date for the demise of Moore's Law keeps getting pushed back. The first estimates predicted 2002, but now Intel says it won't take place until 2022. But as I discussed in chapter 2, every time a specific computing paradigm was seen to approach its limit, research interest and pressure increased to create the next paradigm. This has already happened four times in the century-long history of exponential growth in computation (from electromagnetic calculators to relay-based computers to vacuum tubes to discrete transistors to integrated circuits). We have already achieved many important milestones toward the next (sixth) paradigm of computing: three-dimensional self-organizing circuits at the molecular level. So the impending end of a given paradigm does not represent a true limit.

Highlight (Yellow) | Location 8399

I estimated the capacity of the matter and energy in our solar system to support computation to be at least 1070 cps (see chapter 6). Given that there are at least 1020 stars in the universe, we get about 1090 cps for it, which matches Seth Lloyd's independent analysis. So yes, there are limits, but they're not very limiting.

Highlight (Yellow) | Location 8408

the specific and detailed scenario that I've described by which the software of intelligence will be achieved. The reverse engineering of the human brain, an undertaking that is much further along than Lanier and many other observers realize, will expand our AI toolkit to include the self-organizing methods underlying human intelligence.

Highlight (Yellow) | Location 8415

It is not my intention to defend all software, but it's not true that complex software is necessarily brittle and prone to catastrophic breakdown.

Highlight (Yellow) | Location 8429

functionality always stays on the edge of what's feasible.

Highlight (Yellow) | Location 8430

This romancing of software from years or decades ago is comparable to people's idyllic view of life hundreds of years ago, when people were "unencumbered" by the frustrations of working with machines. Life was unfettered, perhaps, but it was also short, labor-intensive, poverty filled, and disease and disaster prone.

Highlight (Yellow) | Location 8434

In 1985 five thousand dollars bought you a software package that provided a thousand-word vocabulary, did not offer continuous-speech capability, required three hours of training on your voice, and had relatively poor accuracy. In 2000 for only fifty dollars, you could purchase a software package with a hundred-thousand-word vocabulary that provided continuous-speech capability, required only five minutes of training on your voice, had dramatically improved accuracy, offered natural-language understanding (for editing commands and other purposes), and included many other features.⁶

Highlight (Yellow) | Location 8439

I estimate the doubling time of software development productivity to be approximately six years, which is slower than the doubling time for processor price-performance, which is approximately one year today. However, software productivity is nonetheless growing exponentially. The development tools, class libraries, and support systems available today are dramatically more effective than those of decades ago.

Highlight (Yellow) | Location 8442

In my current projects teams of just three or four people achieve in a few months objectives that are comparable to what twenty-five years ago required a team of a dozen or more people working for a year or more.

Highlight (Yellow) | Location 8444

Software Complexity. Twenty years ago software programs typically consisted of thousands to tens of thousands of lines. Today, mainstream programs (for example, supply-channel control, factory automation, reservation systems, biochemical simulation) are measured in millions of lines or more. Software for major defense systems such as the Joint Strike Fighter contains tens of millions of lines.

Highlight (Yellow) | Location 8447

IBM is pioneering the concept of autonomic computing, in which routine information-technology support functions will be automated.⁷ These systems will be programmed with models of their own behavior and will be capable, according to IBM, of being “self-configuring, self-healing, self-optimizing, and self-protecting.” The software to support autonomic computing will be measured in tens of millions of lines of code (with each line containing tens of bytes of information). So in terms of information complexity, software already exceeds the tens of millions of bytes of usable information in the human genome and its supporting molecules.

Highlight (Yellow) | Location 8454

Attempts have been made to formulate measures of software complexity—for example, the Cyclomatic Complexity Metric, developed by computer scientists Arthur Watson and Thomas McCabe at the National Institute of Standards and Technology.⁸ This metric measures the complexity of program logic and takes into account the structure of branching and decision points. The anecdotal evidence strongly suggests rapidly increasing complexity if measured by these indexes, although there is insufficient data to track doubling times.

Highlight (Yellow) | Location 8459

We can already handle levels of software complexity that exceed what is needed to model and simulate the parallel, self-organizing, fractal algorithms that we are discovering in the human brain.

Highlight (Yellow) | Location 8467

to find patterns in signals it is often necessary to solve what are called partial differential equations. Algorithms expert Jon Bentley has shown a continual reduction in the number of computing operations required to solve this class of problem.¹⁰ For example, from 1945 to 1985, for a representative application (finding an elliptic partial differential solution for a three-dimensional grid with sixty-four elements on each side), the number of operation counts has been reduced by a factor of three hundred thousand. This is a 38 percent increase in efficiency each year (not including hardware improvements).

Highlight (Yellow) | Location 8472

the ability to send information on unconditioned phone lines, which has improved from 300 bits per second to 56,000 bps in twelve years, a 55 percent annual increase.¹¹ Some of this improvement was the result of improvements in hardware design, but most of it is a function of algorithmic innovation.

Highlight (Yellow) | Location 8485

The brain uses massive parallel processing as one strategy to achieve more complex functions and faster reaction times, and we will need to utilize this approach in our machines to achieve optimal computational densities.

Highlight (Yellow) | Location 8489

Software improvements, on the other hand, are less predictable. Richards and Shaw call them “worm-holes in development time,” because we can often achieve the equivalent of years of hardware improvement through a single algorithmic improvement.

Highlight (Yellow) and Note | Location 8494

there is a specific game plan for achieving human-level intelligence in a machine: reverse engineer the parallel, chaotic, self-organizing, and fractal methods used in the human brain and apply these methods to modern computational hardware. Having tracked the exponentially increasing knowledge about the human brain and its methods (see chapter 4), we can expect that within twenty years we will have detailed models and simulations of the several hundred information-processing organs we collectively call the human brain.

not clear to me this is what is being done @microsoft research...

Highlight (Yellow) | Location 8498

Many of these methods used extensively in our machine pattern-recognition systems exhibit subtle and complex behaviors that are not predictable by the designer. Self-organizing methods are not an easy shortcut to the creation of complex and intelligent behavior, but they are one important way the complexity of a system can be increased without incurring the brittleness of explicitly programmed logical systems.

Highlight (Yellow) | Location 8501

the human brain itself is created from a genome with only thirty to one hundred million bytes of useful, compressed information. How is it, then, that an organ with one hundred trillion connections can result from a genome that is so small? (I estimate that just the interconnection data alone needed to characterize the human brain is one million times greater than the information in the genome.)¹³ The answer is that the genome specifies a set of processes, each of which utilizes chaotic methods (that is, initial randomness, then self-organization) to increase the amount of information represented.

Highlight (Yellow) | Location 8516

There is nothing wrong with indirect methods; we use them in engineering all the time. Indeed, indirection is how evolution works (that is, the products of each stage create the next stage).

Highlight (Yellow) | Location 8523

chaotic computing, which is how we do pattern recognition, which in turn is the heart of human intelligence. Chaos is part of the process of pattern recognition—it drives the process—and there is no reason that we cannot harness these methods in our machines just as they are utilized in our brains.

Highlight (Yellow) | Location 8527

the essential nature of an evolutionary process: it accelerates because each stage introduces more powerful methods for creating the next stage. We've gone from billions of years for the first steps of biological evolution (RNA) to the fast pace of technological evolution today.

Highlight (Yellow) | Location 8534

the frustration of the AI practitioner: once an AI goal is achieved, it is no longer considered as falling within the realm of AI and becomes instead just a useful general technique. AI is thus often regarded as the set of problems that have not yet been solved.

Highlight (Yellow) | Location 8543

Once we have achieved complete models of human intelligence, machines will be capable of combining the flexible, subtle human levels of pattern recognition with the natural advantages of machine intelligence, in speed, memory capacity, and, most important, the ability to quickly share knowledge and skills.

Highlight (Yellow) | Location 8556

Analog processes can be emulated with digital methods (by using floating point representations), whereas the reverse is not necessarily the case.

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as we have seen our brains are created from a genome of only about thirty to one hundred million bytes of unique information (eight hundred million bytes without compression, but compression is clearly feasible given the massive redundancy), of which perhaps two thirds describe the principles of operation of the brain. It is self-organizing processes that incorporate significant elements of randomness (as well as exposure to the real world) that enable so relatively small an amount of design information to be expanded to the thousands of trillions of bytes of information represented in a mature human brain.

Highlight (Yellow) | Location 8583

the task of creating human-level intelligence in a nonbiological entity will involve creating not a massive expert system comprising billions of rules or lines of code but rather a learning, chaotic, self-organizing system, one that is ultimately biologically inspired.

Highlight (Yellow) | Location 8587

The purpose of reverse engineering the human brain is not to copy the digestive or other unwieldy processes of biological neurons but rather to understand their key information-processing methods.

Highlight (Yellow) | Location 8610

there are no barriers to our discovering the brain's principles of operation and successfully modeling and simulating them, from its molecular interactions upward.

Highlight (Yellow) | Location 8614

The fact that software in a computer is separate from its physical instantiation is an architectural advantage in that it allows the same software to be applied to ever-improving hardware. Computer software, like the brain's changing circuits, can also modify itself, as well as be upgraded.

Highlight (Yellow) | Location 8640

the principles of the design of the brain are simpler than they appear. To understand this, let's first consider the fractal nature of the brain's organization, which I discussed in chapter 2. A fractal is a rule that is iteratively applied to create a pattern or design. The rule is often quite simple, but because of the iteration the resulting design can be remarkably complex.

Highlight (Yellow) | Location 8645

As we look at finer and finer detail in an image of the Mandelbrot set, the complexity never goes away, and we continue to see ever finer complication. Yet the formula underlying all of this complexity is amazingly simple: the Mandelbrot set is characterized by a single formula $Z = Z^2 + C$, in which Z is a "complex" (meaning two-dimensional) number and C is a constant. The formula is iteratively applied, and the resulting two-dimensional points are graphed to create the pattern.

Highlight (Yellow) | Location 8655

it would take at least thousands of trillions of bytes to characterize the state of a human brain, but the design is only tens of millions of bytes.

Highlight (Yellow) | Location 8656

the apparent complexity of the brain to the design information is at least one hundred million to one.

Highlight (Yellow) | Location 8661

(certainly far more complex than the six characters in the definition of the Mandelbrot set), but it is a level of complexity that we can already manage with our technology. Many observers are confused by the apparent complexity in the brain's physical instantiation, failing to recognize that the fractal nature of the design means that the actual design information is far simpler than what we see in the brain.

Highlight (Yellow) | Location 8665

There is, for example, very little information in the genome describing the wiring pattern for the cerebellum, which comprises more than half the neurons in the brain. A small number of genes describe the basic pattern of the four cell types in the cerebellum and then say in essence, "Repeat this pattern several billion times with some random variation in each repetition."

Highlight (Yellow) | Location 8675

unable to simulate the brain on a computer. Because we can describe the brain's principles of operation in mathematical terms, and since we can model any mathematical process (including chaotic ones) on a computer, we are able to implement these types of simulations.

Highlight (Yellow) | Location 8678

Despite his skepticism Bell expresses cautious confidence that we will understand our biology and brains well enough to improve on them. He writes: "Will there be a transhuman age? For this there is a strong biological precedent in the two major steps in biological evolution. The first, the incorporation into eukaryotic bacteria of prokaryotic symbiotes, and the second, the emergence of multicellular life-forms from colonies of eukaryotes.... I believe that something like [a transhumanist age] may happen."

Highlight (Yellow) | Location 8688

As I discussed, quantum computing is computing using what are called qubits, which take on all possible combinations of solutions simultaneously. The method can be considered to be an extreme form of parallel processing (because every combination of values of the qubits is tested simultaneously).

Highlight (Yellow) | Location 8734

Early in the twentieth century mathematicians Alfred North Whitehead and Bertrand Russell published their seminal work, *Principia Mathematica*, which sought to determine axioms that could serve as the basis for all of mathematics.

Highlight (Yellow) | Location 8738

in the 1930s a young Czech mathematician, Kurt Gödel, stunned the mathematical world by proving that within such a system there inevitably exist propositions that can be neither proved nor disproved.

Highlight (Yellow) | Location 8740

Gödel's incompleteness theorem, which is fundamentally a proof demonstrating that there are definite limits to what logic, mathematics, and by extension computation can do, has been called the most important in all mathematics, and its implications are still being debated.²⁷

Highlight (Yellow) | Location 8745

unsolvable problems—that is, problems that are well defined, with unique answers that can be shown to exist, but that we can also show can never be computed by a Turing machine.

Highlight (Yellow) | Location 8754

Taken together, the works of Turing, Church, and Gödel were the first formal proofs that there are definite limits to what logic, mathematics, and computation can do.

Highlight (Yellow) | Location 8757

The weak interpretation is that if a problem that can be presented to a Turing machine is not solvable by one, then it is not solvable by any machine. This conclusion follows from Turing's demonstration that the Turing machine could model any algorithmic process. It is only a small step from there to describe the behavior of a machine as following an algorithm. The strong interpretation is that problems that are not solvable on a Turing machine cannot be solved by human thought, either.

Highlight (Yellow) | Location 8764

although the existence of Turing's unsolvable problems is a mathematical certainty, the Church-Turing thesis is not a mathematical proposition at all. It is, rather, a conjecture that, in various disguises, is at the heart of some of our most profound debates in the philosophy of mind.³⁰

Highlight (Yellow) | Location 8802

Our most complex software (operating systems and telecommunications control systems) already contains tens of millions of lines of code. At present it seems unlikely that we can produce and manage software with hundreds of millions or billions of lines of code.³²

Highlight (Yellow) | Location 8829

Once we can create virtually any physical product from information and very inexpensive raw materials, these traditionally slow-moving industries will see the same kind of annual doubling of price-performance and capacity that we see in information technologies. Energy and transportation will effectively become information technologies.

Highlight (Yellow) | Location 8836

It will be feasible, therefore, to build inexpensive small flying devices that can transport a package directly to your destination in a matter of hours without going through intermediaries such as shipping companies.

Highlight (Yellow) | Location 8841

Because we do not understand the brain very well we are constantly tempted to use the latest technology as a model for trying to understand it. In my childhood we were always assured that the brain was a telephone switchboard. (“What else could it be?”)

Highlight (Yellow) | Location 9032

objective measurement is incompatible with the very concept of subjective experience.

Highlight (Yellow) | Location 9033

It is apparent to me that dogs and cats are conscious (and Searle has said that he acknowledges this as well). But not all humans accept this.

Highlight (Yellow) | Location 9043

we have discovered a biological feature unique to humans and a few other primates: the spindle cells. And these cells with their deep branching structures do appear to be heavily involved with our conscious responses, especially emotional ones. Is the spindle cell structure the neurophysiological basis “x” for human consciousness? What sort of experiment could possibly prove that? Cats and dogs don’t have spindle cells. Does that prove that they have no conscious experience?

Highlight (Yellow) | Location 9081

Drugs are essentially an information technology, and we see the same doubling of price-performance each year as we do with other forms of information technology such as computers, communications, and DNA base-pair sequencing.

Highlight (Yellow) | Location 9096

Human life without death would be something other than human; consciousness of mortality gives rise to our deepest longings and greatest accomplishments. —LEON KASS, CHAIR OF THE PRESIDENTIAL COMMISSION ON BIOETHICS, 2003

Highlight (Yellow) | Location 9099

The criticism concerning governmental control is that regulation will slow down and stop the acceleration of technology. Although regulation is a vital issue, it has actually had no measurable effect on the trends discussed in this book, which have occurred with extensive regulation in place.

Highlight (Yellow) | Location 9114

the Council of Europe proclaimed that “human rights imply the right to inherit a genetic pattern that has not been artificially changed.”⁴¹ Perhaps the most interesting aspect of the council’s edict is its posing a restriction as a right. In the same spirit, I assume the council would advocate the human right not to be cured from natural disease by unnatural means, just as activists “protected” starving African nations from the indignity of consuming bioengineered crops.⁴²

Highlight (Yellow) | Location 9118

Ultimately the benefits of technical progress overwhelm such reflexive antitechnology sentiments.

Highlight (Yellow) and Note | Location 9125

Early in the second decade of this century, the Web will provide full immersion visual-auditory virtual reality with images written directly to our retinas from our eyeglasses and lenses and very high-bandwidth wireless Internet access woven in our clothing. These capabilities will not be restricted just to the privileged. Just like cell phones, by the time they work well they will be everywhere. In the 2020s we will routinely have nanobots in our bloodstream keeping us healthy and augmenting our mental capabilities.

ok... so we're running behind schedule. This is the trouble with saying when in addition to what...

Highlight (Yellow) | Location 9131

Once nonbiological intelligence gets a foothold in our brains, it will at least double in capability each year, as is the nature of information technology. Thus it will not take long for the nonbiological portion of our intelligence to predominate. This will not be a luxury reserved for the rich, any more than search engines are today. And to the extent that there will be a debate about the desirability of such augmentation, it’s easy to predict who will win, since those with enhanced intelligence will be far better debaters.

Highlight (Yellow) | Location 9135

The Unbearable Slowness of Social Institutions. MIT senior research scientist Joel Cutcher-Gershenfeld writes: “Just looking back over the course of the past century and a half, there have been a succession of political regimes where each was the solution to an earlier dilemma, but created new dilemmas in the subsequent era. For example, Tammany Hall and the political patron model were a vast improvement over the dominant system based on landed gentry—many more people were included in the political process. Yet, problems emerged with patronage, which led to the civil service model—a strong solution to the preceding problem by introducing the meritocracy. Then, of course, civil service became the barrier to innovation and we move to reinventing government. And the story continues.”⁴³ Gershenfeld is pointing out that social institutions even when innovative in their day become “a drag on innovation.”

Highlight (Yellow) | Location 9152

There has always been a mix of early and late adopters of new paradigms. We still have people today who want to live as we did in the seventh century.

Highlight (Yellow) | Location 9154

A few hundred years ago, only a handful of people such as Leonardo da Vinci and Newton were exploring new ways of understanding and relating to the world. Today, the worldwide community that participates in and contributes to the social innovation of adopting and adapting to new technological innovation is a substantial portion of the population, another reflection of the law of accelerating returns.

Highlight (Yellow) | Location 9193

It is my thesis that by sharing the complexity as well as the actual patterns of human brains, these future nonbiological entities will display the intelligence and emotionally rich reactions (such as “aspirations”) of humans.

Highlight (Yellow) | Location 9201

Dembski writes: “We need to transcend ourselves to find ourselves.”

Highlight (Yellow) | Location 9204

The material world inherently evolves, and each stage transcends the stage before it. As I discussed in chapter 7, evolution moves toward greater complexity, greater elegance, greater knowledge, greater intelligence, greater beauty, greater creativity, greater love. And God has been called all these things, only without any limitation: infinite knowledge, infinite intelligence, infinite beauty, infinite creativity, and infinite love.

Highlight (Yellow) | Location 9358

I do not know what I may appear to the world, but to myself I seem to have been only like a boy playing on the seashore, and diverting myself in now and then finding a smoother pebble or a prettier shell than ordinary, whilst the great ocean of truth lay undiscovered before me. —ISAAC NEWTON¹

Highlight (Yellow) | Location 9369

In mathematics a singularity is a value that is beyond any limit—in essence, infinity. (Formally the value of a function that contains such a singularity is said to be undefined at the singularity point, but we can show that the value of the function at nearby points exceeds any specific finite value).²

Highlight (Yellow) | Location 9373

With the reverse engineering of the human brain we will be able to apply the parallel, self-organizing, chaotic algorithms of human intelligence to enormously powerful computational substrates. This intelligence will then

be in a position to improve its own design, both hardware and software, in a rapidly accelerating iterative process.

Highlight (Yellow) | Location 9379

A universe saturated with intelligence at 1090 cps would be one trillion trillion trillion trillion times more powerful than all biological human brains on Earth today.³ Even a one-kilogram “cold” computer has a peak potential of 1042 cps, as I reviewed in chapter 3, which is ten thousand trillion (10¹⁶) times more powerful than all biological human brains.⁴

Highlight (Yellow) | Location 9386

If we gained the ability to create and colonize other universes (and if there is a way to do this, the vast intelligence of our future civilization is likely to be able to harness it), our intelligence would ultimately be capable of exceeding any specific finite level. That’s exactly what we can say for singularities in mathematical functions.

Highlight (Yellow) | Location 9410

Our ability to create models—virtual realities—in our brains, combined with our modest-looking thumbs, has been sufficient to usher in another form of evolution: technology. That development enabled the persistence of the accelerating pace that started with biological evolution. It will continue until the entire universe is at our fingertips.

Highlight (Yellow) | Location 9413

Singularity.com

Highlight (Yellow) | Location 9421

KurzweilAI.net

Highlight (Yellow) | Location 9431

Fantastic-Voyage.net and RayandTerry.com For those of you who would like to optimize your health today, and to maximize your prospects of living long enough to actually witness and experience the Singularity, visit Fantastic-Voyage.net and RayandTerry.com. I developed these sites with Terry Grossman, M.D., my health collaborator and coauthor of *Fantastic Voyage: Live Long Enough to Live Forever*.

Highlight (Yellow) | Location 9436

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